PAW3950DM-T5QU: Optical Gaming Navigation Chip

General Description

The PAW3950DM-T5QU is PixArt Imaging's new leading-edge generation of low power, high-end gaming navigation chip, which features a track on glass capability. It is integrated with an illumination source in a 16-pin molded lead-frame DIP package. The chip is designed to fulfill professional gamers' need for a best-in-class gaming experience while enabling casual gaming on glass with its auto-adaptive surface recognition feature. It integrates the enhanced features of high speed, high resolution, high accuracy, selectable lift detection height, and wide surface coverage. The chip is designed with LM19-LSI or L0AE-LSI1 to achieve optimum performance.

Key Features

- Low power consumption at 1.3 mA* (typical) in Run state
- High speed motion detection 750ips* and acceleration 50g*
- Selectable resolutions up to 30,000 cpi with 50cpi step size
- Resolution error of 0.2% (typical) at 5000cpi on QCK up to 200ips*
- Auto-switching between Gaming surface and Glass surface
- Enhanced Programmability
  - High Performance Gaming and Glass Mode
  - Low Power Gaming and Glass Mode
  - Corded Gaming and Glass Mode
  - Office and Office Glass Mode
- Lift detection options
  - Universal Lift Cut-off (minimum 0.7mm on QCK)
  - Manual lift Cut-off calibration
- 16-pin molded lead-frame DIP package with 850nm illumination source
- Angle snapping and Angle tunability
- Self–adjusting variable frame rate for optimum performance
- Internal oscillator — no clock input needed
- 4-wire serial port interface (SPI)
- Motion interrupt output

Applications

- Corded and Cordless optical gaming mice
- Integrated input devices

Key Parameters

Parameter	Value
Supply Voltage	VDD: 1.8 to 2.1 V VDDIO: 1.8 to 3.6 V
Lens Magnification	1:1
Interface	4-wire Serial Port Interface
Typical Operating Current @ VDD = 1.9V, Gaming Mode **Note:** includes LED current	Run: 1.3 mA* (HP Mode) Run: 1.0 mA (LP Mode) Rest1: 550 µA Rest2: 11 µA Rest3: 6 µA Power Down: 4 µA
Resolution	Up to 30,000 cpi
Tracking Speed	750 ips*
Acceleration	50 g*
Package Size	10.90 x 16.20 x 5.01 mm^3
Assembled Solution Size (Package Assemble with lens)	LM19-LSI: 18.85 x 21.15 x 9.81mm^3 L0AE-LSI1: 10.90 x 16.80 x 9.81mm^3

Note: * High Performance Gaming Mode

Ordering Information

Part Number	Description	Package Type	Packing Type	MOQ
PAW3950DM-T5QU	Optical Gaming Navigation Chip	16-pin DIP	Tube	1000
LM19-LSI	Lens	Plastic lens	Tray	1000
L0AE-LSI1	Trim Lens	Plastic lens	Tray	1000

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1.0 Introduction

1.1 Overview

The PAW3950DM-T5QU contains a Picture Element Acquisition System (PEAS), a hard-coded Digital Signal Processing System (DSPS), and a four-wire serial port interface. It is based on Optical Navigation Technology to measure changes in a position where acquiring sequential picture elements for calculating the direction and magnitude of movement (delta X and delta Y).

Note: Throughout this document, the PAW3950DM-T5QU is referred to as the "chip".

1.2 Block Diagram

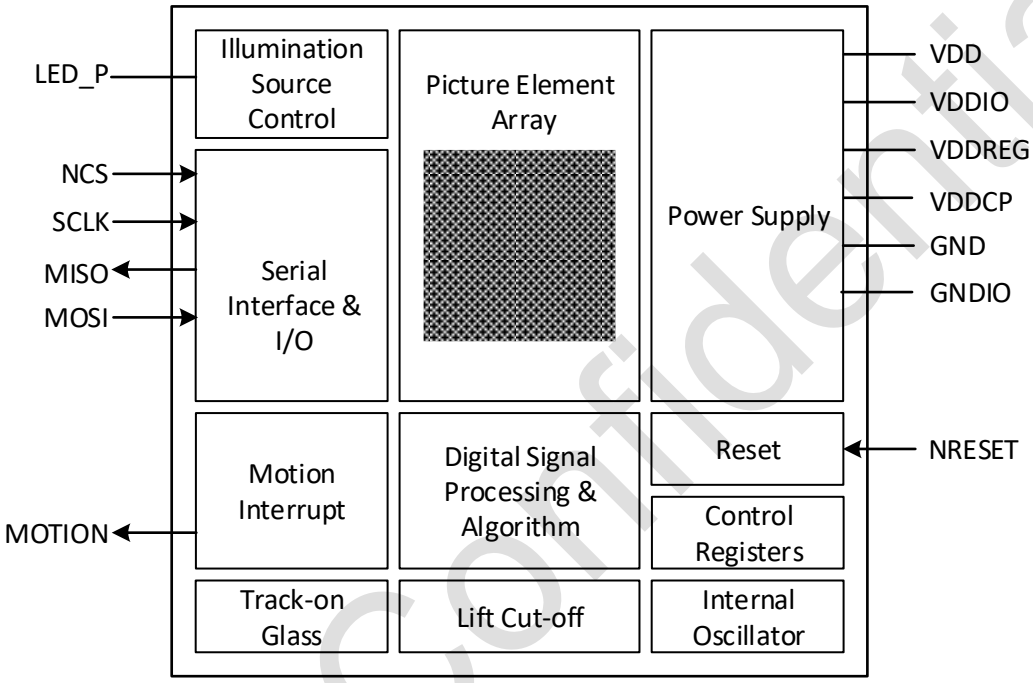


Figure 1. Block Diagram

1.3 Terminology

Term	Description
DSPS	Digital Signal Processing System
ESD	Electrostatic Discharge
LED	Light Emitting Diode
IC	Integrated Circuit
I/O	Input / Output
IR	Infrared
PCB	Printed Circuit Board
TOG	Track on Glass

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1.4 Pin Assignment and Signal Description

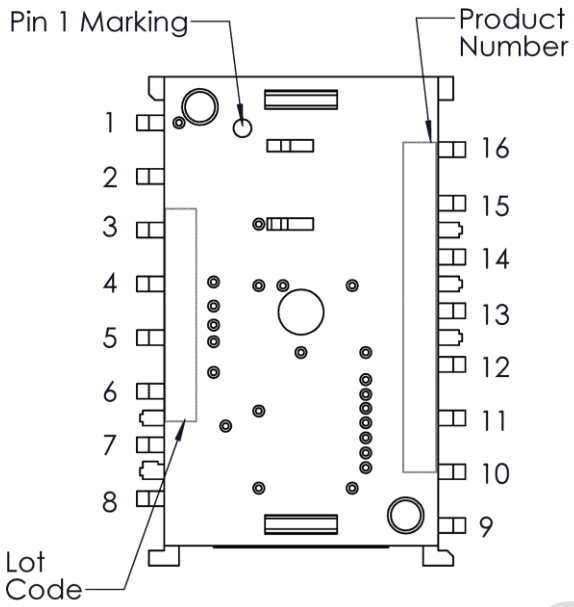


Figure 2. Pin Configuration

Table 1. Signal Pins Description

Pin No.	Signal Name	Type	Description
1	NC	NC	No connection
2	VDDCP	Power	Charge pump output
3	GND	Ground	Ground
4	VDD	Power	Input power supply
5	VDDREG	Power	LDO output for digital core (only for internal usage)
6	NC	NC	No connection
7	VDDIO	Power	I/O power supply
8	GNDIO	Ground	I/O Ground
9	MOTION	Output	Motion detection
10	SCLK	Input	4-wire SPI Serial data clock
11	MOSI	Input	4-wire SPI Serial data input
12	MISO	Output	4-wire SPI Serial data output
13	NCS	Input	4-wire SPI Chip select (Active Low)
14	NRESET	Input	Chip reset (Active Low)
15	LED_P	Input	LED Anode
16	NC	NC	No connection

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2.0 Operating Specifications

2.1 Regulatory Requirement

- Passes FCC Part15, Subpart B, Class B, ANSI C63.4:2014. Tested with a reference design mouse with shielded High-Speed USB 2.0 cable with ferrite bead under PixArt's recommended operating conditions.
- Passes IEC 62471: 2008 Photo biological safety of lamps and lamp systems.

2.2 Absolute Maximum Ratings

Table 2. Absolute Maximum Ratings

Parameter	Symbol	Min.	Max.	Unit	Note
Storage Temperature	T_S	-40	85	°C	
Supply Voltage	VDD	-0.5	2.1	V	
	VDDIO	-0.5	3.6	V	
I/O Pin Voltage		-0.5	VDDIO	V	All I/O pins
Lead-Free Solder Temperature	T_{SOLDER}		260	°C	For 7 seconds, 1.6mm below seating plane
ESD	ESD_{HBM}		2	kV	All pins (Human Body Model)

Notes:

1. Maximum Ratings are the maximum parameter values that can damage the device when exceeding this limit.
2. Exposure to these conditions or conditions beyond those indicated may adversely affect device reliability. Functional operation under absolute-maximum-rated conditions is not recommended.

2.3 Recommended Operating Conditions

Table 3. Recommended Operating Conditions

Parameter		Symbol	Min.	Typ.	Max.	Unit	Note
Operating Temperature		T_A	0		40	°C	
Power Supply Voltage		VDD	1.8	1.9	2.1	V	Including supply noise
		VDDIO	1.8	1.9	3.6		Including supply noise. (VDDIO must be equal or greater than VDD)
Power Supply Rise Time		t_{RT}	0.15		20	ms	0 to VDD minimum
Supply Noise (peak to peak)		V_{NA}			100	mV	10 kHz to 75 MHz
Serial Port Clock Frequency		f_{SCLK}			16	MHz	50% duty cycle
Distance from Lens Reference Plane to Navigation Surface		Z	2.2	2.4	2.6	mm	
Gaming Mode	Resolution	Res			30000	cpi	High Performance, Low Power and Corded Setting
					16000		On Glass surfaces, good tracking performance up to 16000cpi. Maximum resolution can be configured up to 30000cpi
	Maximum Speed	S	750			ips	High Performance Setting

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	Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
	In Run State at 45 degree		550				Low Power Setting
			750				Corded Setting
			80				Glass
	Maximum Acceleration In Run State at 45 degree	A	50			g	High Performance Setting
			40				Low Power Setting
			50				Corded Setting
			15				Glass
	Resolution Error Up to 200ips on QCK at 5000 cpi	Res_{Err}		0.2		%	High Performance Setting
				0.3			Low Power Setting
				0.2			Corded Setting
	Resolution Error Up to 80 ips on glass surface at 1000 cpi	Res_{Err}		2.0		%	Glass
	Lift Cut-off On PixArt's standard gaming surfaces	Lift		0.7		mm	Lift Setting A
				1.0			Lift Setting B
				2.0			Lift Setting C
	Lift Cut-off On Glass	Lift		1.0		mm	Lift Setting A
				1.0			Lift Setting B
				2.0			Lift Setting C
Office Mode	Resolution	Res			30000	cpi	PixArt's standard gaming surfaces
					16000		On Glass surfaces, good tracking performance up to 16000cpi. Maximum resolution can be configured up to 30000cpi
	Maximum Speed In Run State at 45 degree	S	200			ips	PixArt's standard gaming surfaces
			30				On Glass surfaces
	Maximum Acceleration In Run State at 45 degree	A	10			g	
	Lift Cut-off On PixArt's standard gaming surfaces	Lift		0.7		mm	Lift Setting A
				1.0			Lift Setting B
				2.0			Lift Setting C
	Lift Cut-off On Glass Surface	Lift		1.0		mm	Lift Setting A
				1.0			Lift Setting B
				2.0			Lift Setting C

Note:

1. The chip is designed to work on glass (including the transparent glass) with thickness of at least 4mm. Most glass surfaces provide enough microscopic features to enabled motion tracking. It will not work on the glass which is perfectly clean and virtually scratch-free.
2. PixArt does not guarantee the performance of the system beyond the recommended operating condition limits.

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2.4 DC Characteristics

Table 4. DC Electrical Specifications

Parameter		Symbol	Min.	Typ.	Max.	Unit	Note
DC Supply Current (Gaming Mode)	High Performance Setting	IDD_{RUN}		1.3		mA	▪ IDD_{RUN}: Average current consumption up to 200ips, including LED current with 1ms polling ▪ IDD_{REST}: Average current consumption including LED current
		IDD_{REST1}		550		µA	
		IDD_{REST2}		11		µA	
		IDD_{REST3}		6		µA	
	Low Power Setting	IDD_{RUN}		1.0		mA	
		IDD_{REST1}		550		µA	
		IDD_{REST2}		11		µA	
		IDD_{REST3}		6		µA	
	Corded Setting	IDD_{RUN}		13.0		mA	▪ IDD_{RUN}: Average current consumption up to 750ips, including LED current with 0.125ms polling
	Glass	IDD_{RUN}		3.0		mA	▪ IDD_{RUN}: Average up to 80ips ▪ IDD_{RUN}: average current consumption, including LED current with 1ms polling ▪ IDD_{REST}: Average current consumption including LED current
		IDD_{REST1}		450		µA	
		IDD_{REST2}		30		µA	
		IDD_{REST3}		15		µA	
DC Supply Current (Office Mode)	PixArt's Standard Gaming Surfaces	$*^{1}IDD_{RUN}$		0.35		mA	▪ $*^{1}IDD_{RUN}$: average up to 30ips ▪ $*^{2}IDD_{RUN}$: average up to 200ips ▪ IDD_{RUN}: Average current consumption, including LED current with 8ms polling ▪ IDD_{REST}: Average current consumption including LED current
		$*^{2}IDD_{RUN}$		0.50		mA	
		IDD_{REST1}		60		µA	
		IDD_{REST2}		11		µA	
		IDD_{REST3}		6		µA	
	Glass Surfaces	IDD_{RUN}		2.0		mA	▪ IDD_{RUN}: Average current consumption up to 30ips, including LED current with 8ms polling ▪ IDD_{REST}: Average current consumption including LED current
		IDD_{REST1}		450		µA	
		IDD_{REST2}		30		µA	
		IDD_{REST3}		15		µA	
Shutdown Current		I_{STDWN}		4		µA	
Input Hysteresis		V_{I_HYS}		100		mV	SCLK, MOSI, NCS
Input Leakage Current		I_{leak}		±1	±10	µA	V_{in}= VDDIO or 0V, SCLK, MOSI, NCS
Input Low Voltage		V_{IL}			0.3 x VDDIO		SCLK, MOSI, NCS
Input High Voltage		V_{IH}	0.7 x VDDIO				SCLK, MOSI, NCS

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Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Output Low Voltage	V_{OL}			0.45	V	I_{out}= 1mA for MISO I_{out}= 0.1mA for MOTION
Output High Voltage	V_{OH}	VDDIO - 0.45			V	I_{out}= -1mA for MISO I_{out}= -0.1mA for MOTION
Transient Supply Current	I_{DDT}			70	mA	Maximum supply current during the supply ramp from 0V to VDD with minimum 150 µs and maximum 20ms rise time. (Does not include charging currents for bypass capacitors)
	I_{DDTIO}			60	mA	Maximum supply current during the supply ramp from 0V to VDDIO with minimum 150 µs and maximum 20ms rise time. (Does not include charging currents for bypass capacitors)

Note: Chip electrical characteristics over recommended operating conditions. Typical values at 25°C, VDD= 1.9V, VDDIO= 1.9V, and with LED current at 50mA.

2.5 AC Characteristics

Table 5. AC Electrical Specifications

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
Motion Delay After Reset	$t_{MOT-RST}$	60			ms	From reset to valid motion, assuming motion is present
Shutdown	t_{STDWN}			500	ms	From Shutdown state active to low current
Wake from Shutdown	t_{WAKEUP}	60			ms	From Shutdown state inactive to valid motion.

Note: All the parameters are tested under operating conditions: VDD= 1.9V, VDDIO= 1.9V, T_A= 25°C.

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2.6 SPI Specifications

Table 6. SPI Timing Specifications

Parameter	Symbol	Min.	Typ.	Max.	Unit	Note
MISO Rise Time	$t_{r\text{-}MISO}$		5		ns	C_L= 20pF
MISO Fall Time	$t_{f\text{-}MISO}$		5		ns	C_L= 20pF
MISO Delay After SCLK	$t_{DLY\text{-}MISO}$			17	ns	From SCLK falling edge to MISO data valid. C_L= 20pF
MISO Hold Time	$t_{hold\text{-}MISO}$	15.6			ns	Data held until next falling SCLK edge
MOSI Hold Time	$t_{hold\text{-}MOSI}$	15.6			ns	Amount of time data is valid after SCLK rising edge
MOSI Setup Time	$t_{setup\text{-}MOSI}$	15.6			ns	From data valid to SCLK rising edge.
SPI Time Between Write Commands	t_{SWW}	3			µs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second data byte.
SPI Time Between Write And Read Commands	t_{SWR}	2.5			µs	From rising SCLK for last bit of the first data byte, to rising SCLK for last bit of the second address byte.
SPI Time Between Read And Subsequent Commands	t_{SRW} t_{SRR}	2.5			µs	From rising SCLK for last bit of the first data byte, to falling SCLK for the first bit of the address byte of the next command.
SPI Read Address-Data Delay	t_{SRAD}	1			µs	From rising SCLK for last bit of the address byte, to falling SCLK for first bit of data being read.
NCS Inactive After Motion Burst	t_{BEXIT}	300			ns	Minimum NCS inactive time after motion burst before next SPI usage.
NCS To SCLK Active	$t_{NCS\text{-}SCLK}$	120			ns	From last NCS falling edge to first SCLK rising edge.
SCLK To NCS Inactive (For Read Operation)	$t_{SCLK\text{-}NCS}$	120			ns	From last SCLK rising edge to NCS rising edge, for valid MISO data transfer.
SCLK To NCS Inactive (For Write Operation)	$t_{SCLK\text{-}NCS}$	1			µs	From last SCLK rising edge to NCS rising edge, for valid MOSI data transfer.
NCS To MISO High-Z	$t_{NCS\text{-}MISO}$			500	ns	From NCS rising edge to MISO high-Z state.
MOTION Rise Time	$t_{r\text{-}MOTION}$		300		ns	C_L= 20pF
MOTION Fall Time	$t_{f\text{-}MOTION}$		300		ns	C_L= 20pF
Input Capacitance	C_{in}		10		pF	SCLK, MOSI, NCS, NRESET
Load Capacitance	C_L			20	pF	MISO, MOTION

Note: All the parameters are tested under operating conditions: VDD= 1.9V, VDDIO= 1.9V, T_A= 25°C.

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3.0 Mechanical Specifications

3.1 Chip Package Outline Drawing

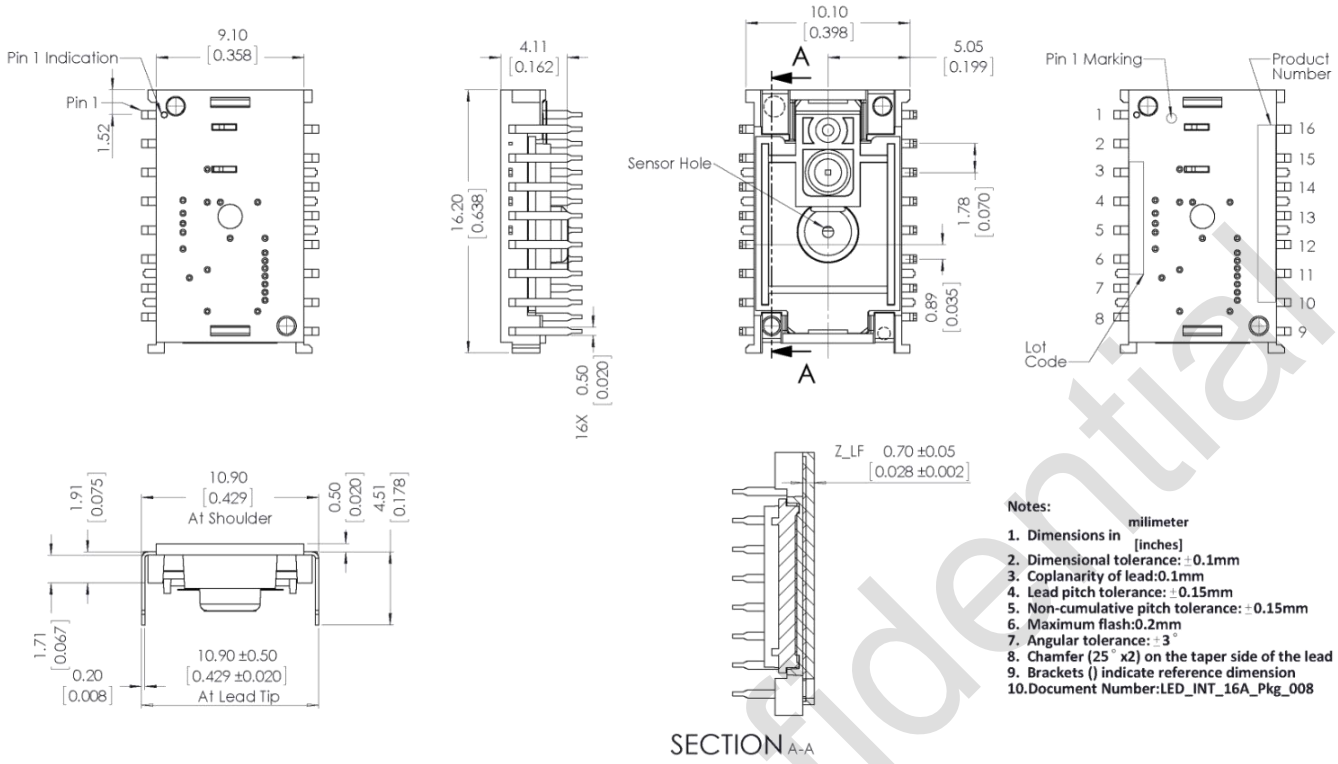


Figure 3. Package Outline Drawing

3.2 Package Marking

Refer to Figure 2. Pin Configuration for the code marking location on the device package.

Table 7. Code Identification

Code	Marking	Description
Product Number	PAW3950DM-T5QU	Part number label
Lot Code	AYWWXXXXX	A: Assembly house Y: Year WW: Week XXXXX: PixArt reference

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3.3 Packing Information

Item	Description
Product number	PAW3950DM-T5QU
Package type	16L DIP
Quantity per tube	25 pcs
Quantity per Inner box	1,000 pcs
Quantity per Shipping box	12,000 pcs
Tube size	500 x 13.5 x 7.0 mm^3
Inner packing box size	163 x 525 x 67 mm^3
Outer packing box size	365 x 560 x 240 mm^3 (≤ 6,000 pcs) 540 x 555 x 300 mm^3 (> 6,000 pcs to 12,000pcs)

3.3.1 Packing Tube

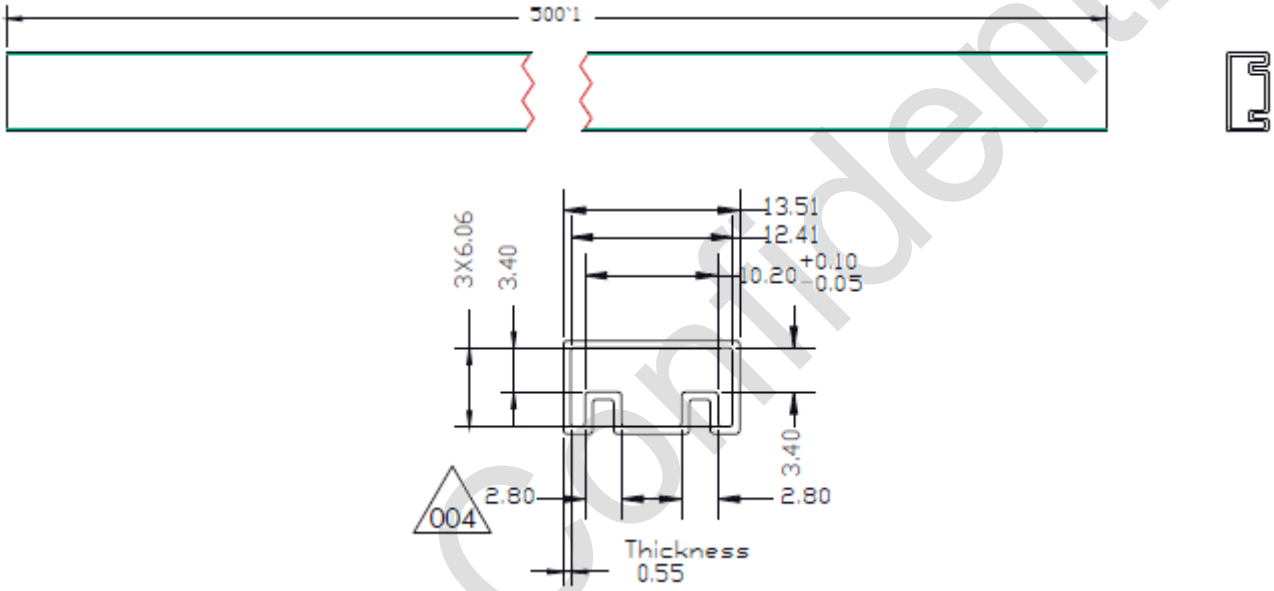


Figure 4. Packing Tube Dimension

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3.4 Package Handling Information

3.4.1 Sample of Inner Box Label



Note: This label is used on inner box

3.4.2 Sample of Shipping Box Label



Note: This label is used on shipping box

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4.0 Design Reference

4.1 General Reference Schematic

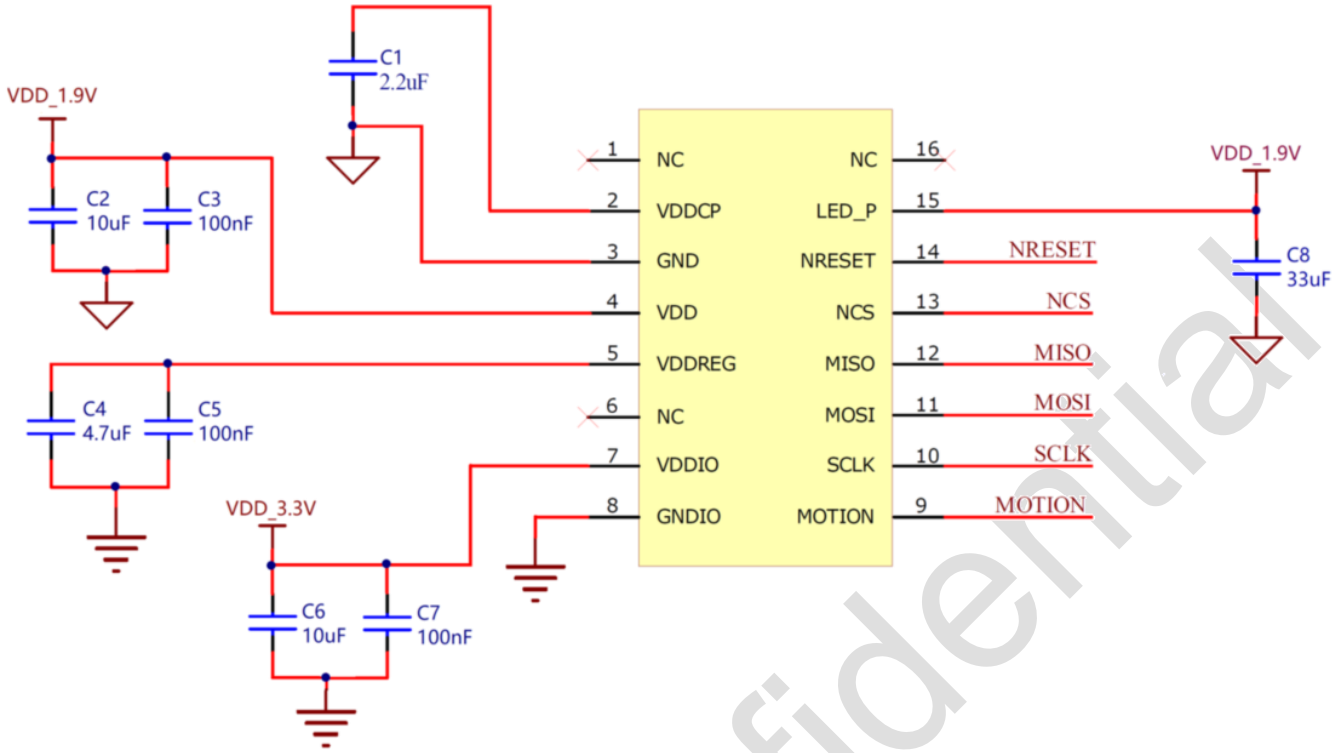


Figure 5. Reference Schematic Diagram

Note:

1. Do not leave the NRESET floating, it must be driven constantly by the host to establish its state.
2. All the capacitors must be placed as close as possible to the pin.
3. NC pins should be left unconnected.
4. Recommended specification for capacitor C1:

Parameter	Specification (Value)
Dielectric Characteristic	X5R
Insulation Resistance	10,000MΩ or 100MΩ × μF Whichever is smaller

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4.2 PCB Layout Design Guide

4.2.1 Recommended PCB Footprint

It is highly recommended to follow the chip orientation in Figure 6 to achieve optimum tracking performance.

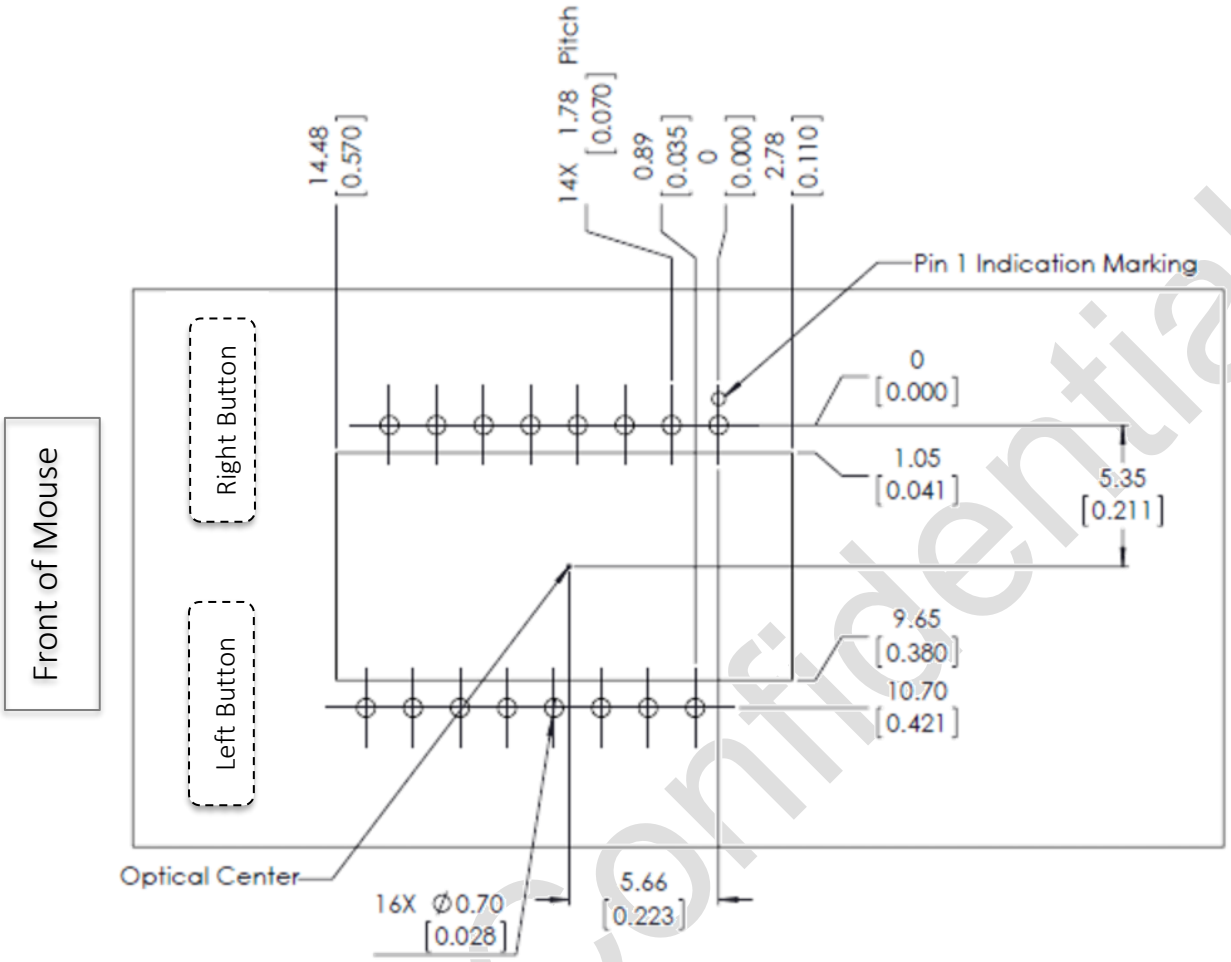


Figure 6. Recommended PCB Footprint

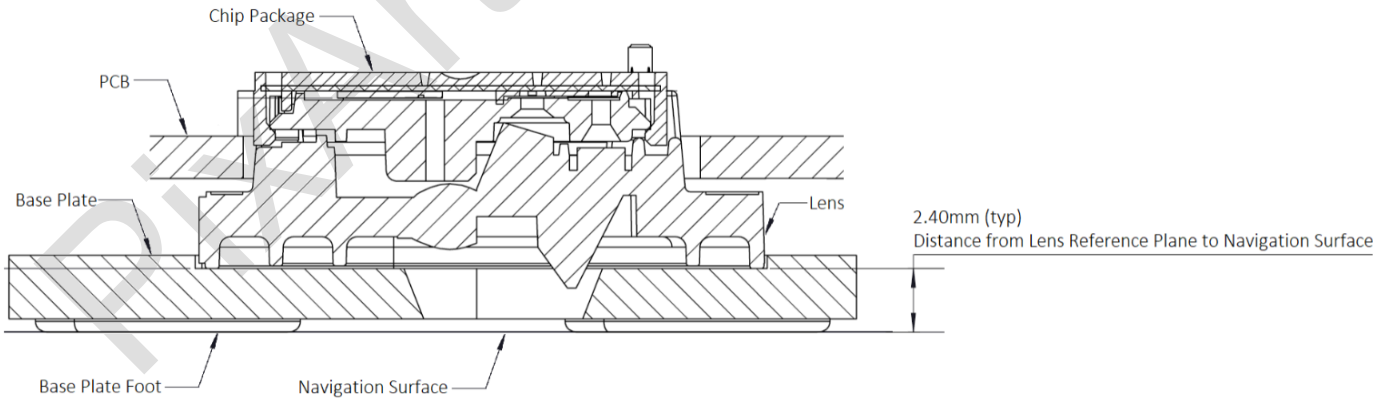


Figure 7. Distance from Lens Reference Plane to Tracking Surface

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4.3 Assembly Guide

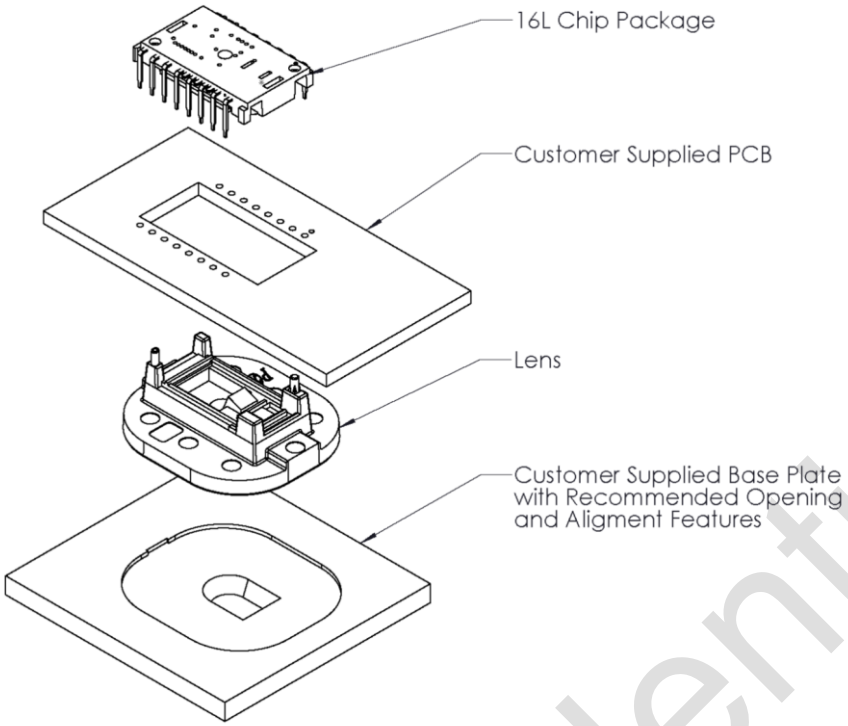
Proper grounding during handling and assembly of the components is required to prevent the chip from damage or performance degradation.

4.3.1 Assembly Recommendation

1. Insert the integrated chip and all other electrical components into PCB. The force applied on the integrated chip package must not have a contact area smaller than 3mm in diameter and should not exceed a maximum static force of 2.5kgf.
2. Wave-solder the entire assembly in a no-wash solder process utilizing solder-fixture. A solder-fixture is required to protect the chip from flux spray and wave solder paste.
3. Avoid getting any solder flux onto the chip body as there is potential for flux to seep into the chip package, the solder fixture must be designed to expose only the chip leads to flux spray and molten solder while shielding the chip body and optical apertures. The fixture must set the chip at the correct position and height on the PCB.
4. Place the lens onto the base plate and avoid contamination on the optical surfaces.
5. Remove the protective Kapton tapes from optical apertures of the chip. Avoid contaminants from entering the apertures. Do not place the PCB with the chip facing up during the entire mouse assembly process. Hold the PCB vertically when removing Kapton tape.
6. Insert PCB assembly over the lens onto the base plate aligning post to retain PCB assembly. The chip package will self-align to the lens via the guide posts. The optical position reference for the PCB is set by the base plate and lens. Note that the PCB motion due to button presses must be minimized to maintain optical alignment.
7. Recommendation: The lens can be permanently secured to the chip package by melting the lens' guide posts over the chip with heat staking process. Please refer to Application Note titled *"LM19-LSI Lens: PCB Assembly & Lens Heat Staking Recommendations"* for details and recommendation on the lens heat staking process.
8. Install mouse top case. There must be a fixture in the top case to press down onto the PCB assembly to ensure all components are stacked or interlocked to the correct vertical height.
9. It is recommended to place mouse feet around the base plate opening to stabilize mouse tracking on the surface.

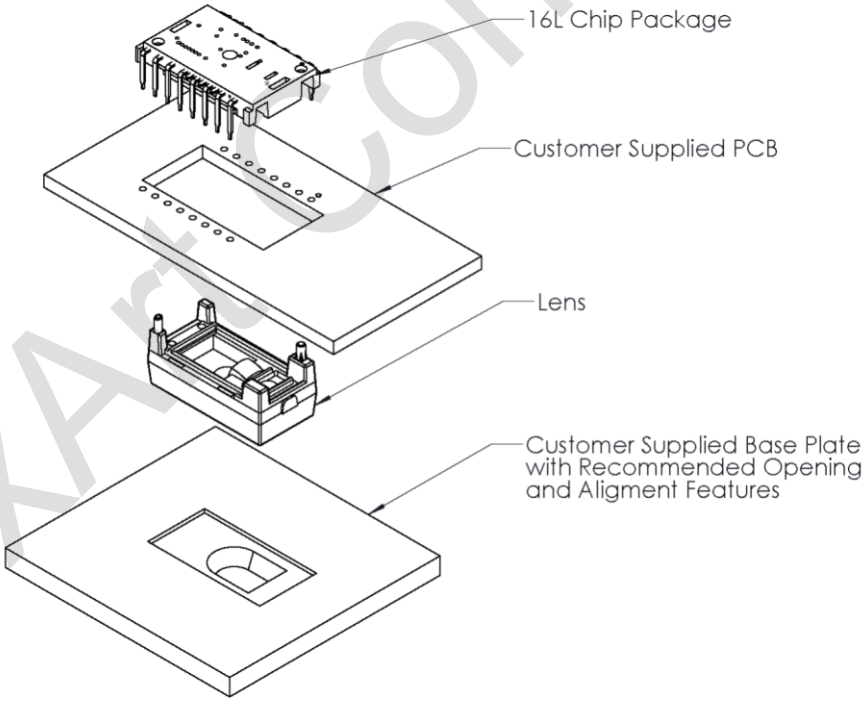
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Note: Refer to LM19-LSI lens datasheet for the detail.

Figure 8. Exploded View of Assembly with LM19-LSI Lens



Note: Refer to L0AE-LSI1 lens datasheet for the detail.

Figure 9. Exploded View of Assembly with L0AE-LSI1

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5.0 Power Management

5.1 Power Supply

The chip has two power supply inputs which are VDD and VDDIO. VDD is the main power supply and VDDIO is the I/O reference voltage. It has two power supply outputs which are charge Pump and LDO output for digital core.

5.2 Power Sequence Requirement

5.2.1 Power-on Sequence

VDD and VDDIO can be powered on in any order with less than 100ms of interval between VDD and VDDIO.

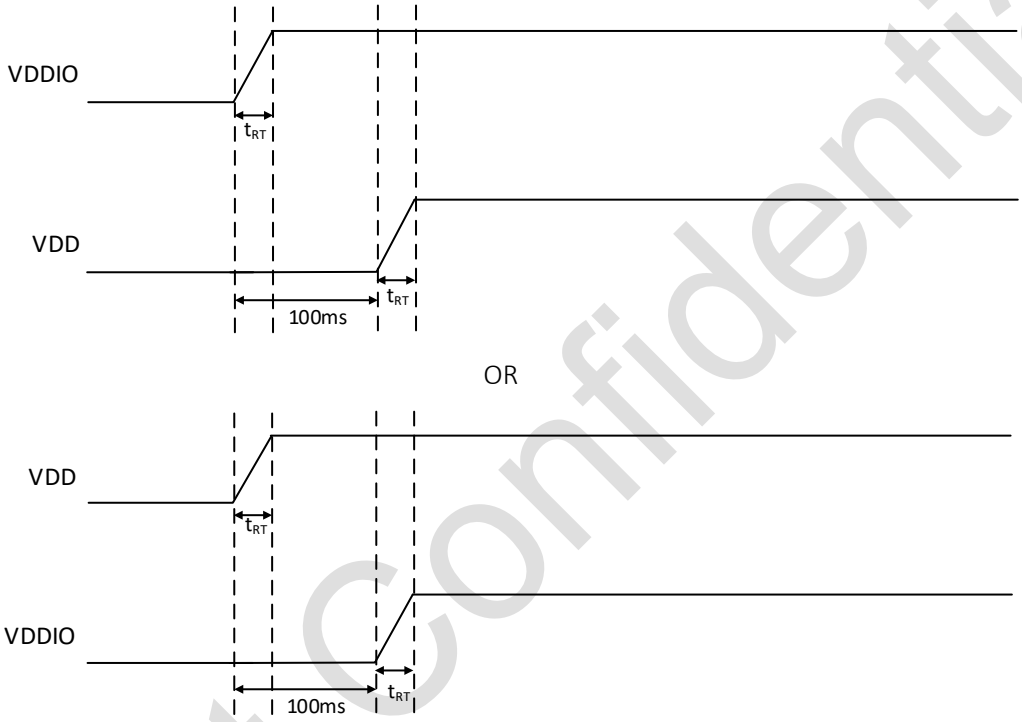


Figure 10. Power-on Sequence Requirement

5.2.2 Power-off Sequence

Power off both VDD and VDDIO at the same time or power-off VDD first, followed by VDDIO.

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5.3 Power State

5.3.1 State Description

Table 8. State Description

State	Description
OFF	No power supply, all the voltage rails and clocks are gated.
RUN	The chip enters RUN state after the regular power on and initialization.
REST 1 REST 2 REST 3	The chip enters REST state if no motion is detected after a period of time for power saving purpose. The chip exits REST state and transition back to RUN state upon motion is detected. REST 1, REST 2 and REST 3 are power-saving states. Each state has a different motion detection period with its respective response time. The current consumption is lowest at Rest 3 and highest at Rest 1. However, the time required for the chip to respond to the motion from Rest 1 is the shortest and longest from Rest 3. This feature can be enabled or disabled through register command.
Shutdown	The chip enters this state upon receive host command for power saving purpose. To exit Shutdown state, only reset command (write 0x5A to the *Power_Up_Reset* register) is accepted. The chip has no motion detection function in this state.

5.3.2 State Diagram

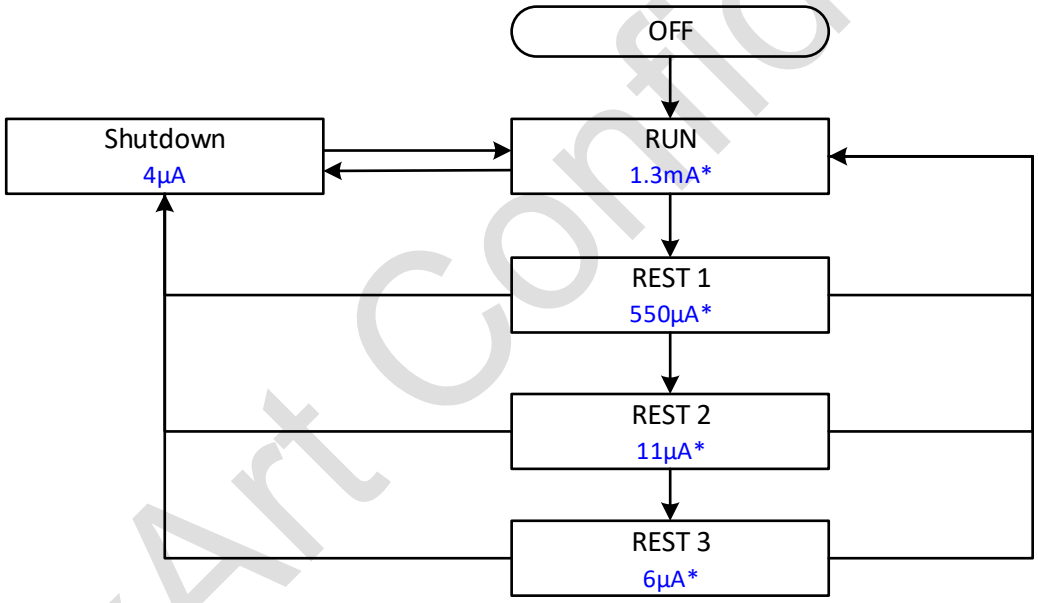


Figure 11. Power State Diagram

Note:

1. *Typical current with “High Performance Gaming Mode” setting
2. Refer to Table 4 for average current consumption across different operation modes.

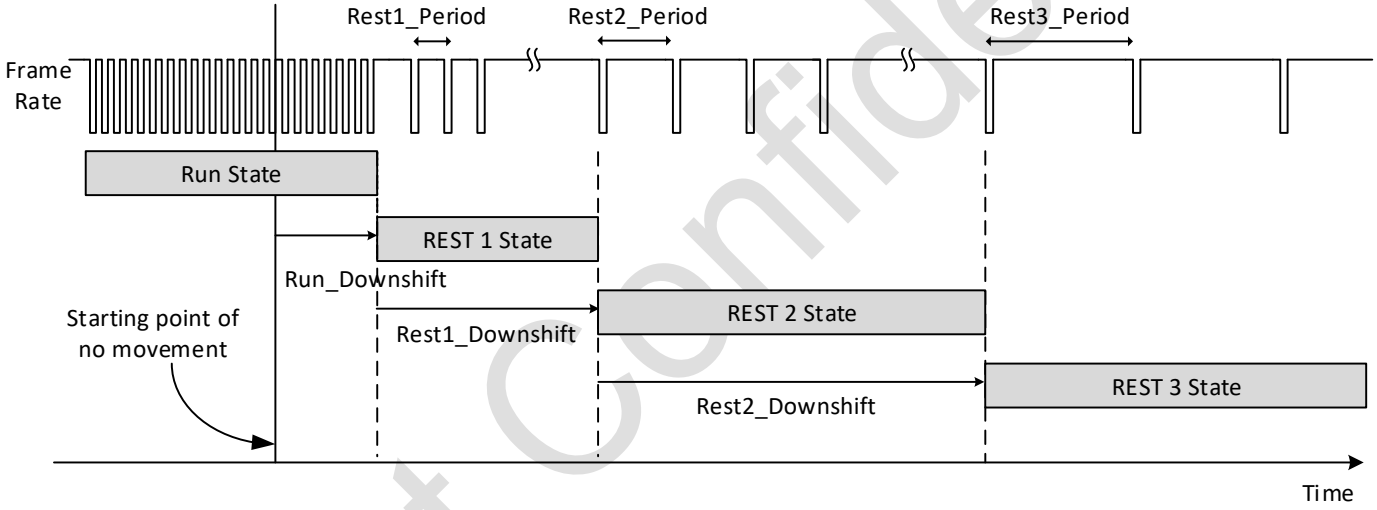
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5.3.3 State Transition

State	Type	Description
OFF to RUN	Power-up	After the chip is powered on and initialization is completed, the chip enters RUN state.
RUN to REST 1 REST 1 to REST 2 REST 2 to REST 3	Timer Counter	After the chip does not detect motion for a period of time, the chip enters REST state automatically. When no motion is detected, the chip gradually downshifts from RUN to REST 1, REST 2 and finally to REST 3 automatically.
REST 1 to RUN REST 2 to RUN REST 3 to RUN	Motion Detection	Upon the chip detects motion during REST 1, REST 2 or REST 3, the chip enters RUN state automatically.
RUN, REST 1, REST 2 or REST 3 to Shutdown	Command	Enter Shutdown state by command.
Shutdown to RUN	Reset	Reset the chip to exit Shutdown state, followed by system initialization. After the chip is initialized, it enters RUN state.

- Run to REST 1, REST 2 and REST 3



The chip supports three power saving states (REST 1, REST 2 and REST 3), each state has its downshift time settings. When no motion is detected, the chip gradually downshifts from Run State to REST 1, REST 2 and finally to REST 3 state automatically. Downshift time is the elapsed time to control the state transition. The default downshift time for each rest state is as shown in Table 9 and Table 10.

Table 9. Gaming Mode and Office Mode Downshift Time

State	Downshift Time		Relevant Register
	HP Mode/ LP Mode	Office Mode	
Run to Rest 1	1s	1s	*Run_Downshift*[7:0], *Run_Downshift_Mult*[3:0]
Rest 1 to Rest 2	10s	10s	*Rest1_Downshift*[7:0], *Rest_Downshift_Mult*[2:0], *Rest1_Period*[7:0]
Rest 2 to Rest 3	600s	600s	*Rest2_Downshit*[7:0], *Rest_Downshift_Mult*[6:4], *Rest2_Period*[7:0]

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Table 10. Glass Mode Downshift Time

State	Downshift Time	Relevant Register
Run to Rest 1	1s	*Run_Downshift*[7:0], *Run_Downshift_Mult*[3:0]
Rest 1 to Rest 2	10s	*Glass_Rest1_Downshift*[7:0], *Rest_downshift_Mult*[2:0], *Glass_Rest1_period[7:0]*
Rest 2 to Rest 3	600s	*Rest2_Downshit*[7:0], *Rest_Downshift_Mult*[6:4], *Rest2_period*[7:0]

- REST 1, REST 2 or REST 3 to Run

When motion is detected, the chip exit from Rest state and transits back to Run state. Response time is the time taken for the chip to wake up from the rest state when motion is detected.

Maximum response time can be calculated as follows:

Max. Response Time = 2 x Rest Period

The default response time for each mode as shown in Table 11 and Table 12. The timings are based on power-up initialization register setting in the section 7.1.2. The timings are subjected to change if any of the values in register 0x74 to 0x75 and 0x77 to 0x7E are updated. Note that changing the rest period will impact to the rest state's current consumption.

Table 11. Gaming Mode and Office Mode Response Time

Rest State	Response Time		Relevant Register
	HP Mode / LP Mode	Office Mode	
Rest 1	< 2ms	< 20ms	*Rest1_Period*[7:0]
Rest 2	< 200ms	< 200ms	*Rest2_Period*[7:0]
Rest 3	< 1000ms	< 1000ms	*Rest3_Period*[7:0]

Table 12. Glass Mode Response Time

Rest State	Response Time	Relevant Register
Rest 1	< 10ms	*Glass_Rest1_Period*[7:0]
Rest 2	< 200ms	*Rest2_Period*[7:0]
Rest 3	< 1000ms	*Rest3_Period*[7:0]

- RUN to Shutdown

Write register 0x3B with value 0xB6 to enter Shutdown state.

- Exit Shutdown

To de-assert Shutdown state and enter RUN state, reset the chip, and perform system initialization in the section 7.1.1, start from step 3.

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5.3.4 Pin State in Shutdown State

In shutdown state, the SPI port must not be accessed during shutdown state except the reset command (write 0x5A to register 0x3A).

It is recommended to de-assert NCS pin during shutdown state.

When NCS is de-asserted:

- MISO is internal weak pull high
- SCLK and MOSI are invalidated

The table below shows the state of various pins during shutdown state.

Table 13. State of Signal Pins during Shutdown State

Pin	Status during Shutdown State
NRESET	High
NCS	High
MISO	Internal weak pull high
SCLK	Ignore if NCS= 1
MOSI	Ignore if NCS= 1
MOTION	Output High

5.4 Reset

There are three approaches to trigger for chip reset. The chip needs to re-initialize due to all previous registers will be cleared after each reset.

The table below shows the state of the various pins during power-up and reset.

Table 14. State of Signal Pins during Power-on and Reset

State of Signal Pins after VDD is Valid		
Pin	During Reset	After Reset
NRESET	Functional	Functional
NCS	Ignored	Functional
MISO	Undefined	Depends on NCS
SCLK	Ignored	Depends on NCS
MOSI	Ignored	Depends on NCS
MOTION	Undefined	Functional

5.4.1 Power-on-Reset

For every power-on, it is recommended to perform either Hardware Reset or Software Reset.

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5.4.2 Hardware Reset

The NRESET pin is used to perform full chip reset. When asserted, it performs the same reset function as the *Power_Up_Reset* register. The NRESET pin needs to be asserted (held to logic 0) for at least 100 ns duration for the chip to reset.

Note: NRESET pin has a built-in weak pull-up circuit. During active low reset phase, the NRESET pin can draw a static current of up to 600µA.

5.4.3 Software Reset

Write 0x5A to the *Power_Up_Reset* register to reset the chip. All register settings will be reverted to default values, the chip is required for re-initialization.

5.5 Related Register

Usage	Name	Bank	Address
Downshift Time and Rest Period	*Glass_Rest1_Period*	0x00	0x74
	Glass_Rest1_Downshift	0x00	0x75
	Run_Downshift	0x00	0x77
	Rest1_Period	0x00	0x78
	Rest1_Downshift	0x00	0x79
	Rest2_Period	0x00	0x7A
	Rest2_Downshift	0x00	0x7B
	Rest3_Period	0x00	0x7C
	Run_Downshift_Mult	0x00	0x7D
	Rest_Downshift_Mult	0x00	0x7E
Power Management	*Shutdown Register*	0x00	0x3B
	Power_Up_Reset	0x00	0x3A

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6.0 Control Interface

The chip adopts Serial Port Interface Communication (SPI) interface. The interface is configured to SPI mode 3 (CPHA= 1 and CPOL= 1) for slave device communication. The maximum SCLK speed is 16MHz.

6.1 Signal Description

The 4-wire synchronous serial port interface is used to write or read registers in the chip. The host is a SPI master device where it drives SCLK, MOSI, and NCS (active low) to initiate the communication. The interface supports single or multi-chip (slaves) with NCS control.

Table 15. Signal Description

Pin	Reset Status	Description
SCLK	High	Clock input, generated by the host (master).
MOSI	High	Input data (Master Out / Slave In).
MISO	Weak Pull High	Output data (Master In / Slave Out).
NCS	High	Chip select input (active low).

6.2 Chip Select Operation

The SPI port is activated after NCS is asserted. If NCS is de-asserted during a transaction, the entire transaction is aborted and the serial port will be reset. After a transaction is aborted, the host needs to follow the normal address-to-data or transaction-to-transaction delay timing requirement before beginning the next transaction.

When the NCS pin is de-asserted, the MOSI are ignored and the MISO are in weak pull-high. NCS can also be used to reset the serial port communication in case of an error occur.

To improve communication reliability, all serial transactions must be framed by NCS assertion. The port must not be remained enabled when not in use to prevent misinterpret ESD and EFT/B events and put the chip into an unknown state.

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6.3 Protocol

The transmission protocol is a 4-wire link, half duplex protocol between the host and the chip. All data changes at SCLK falling edge and latched at SCLK rising edge. The host controller always initiates communication and the chip never initiates data transfers.

The transmission protocol consists of the two type operations:

- Write Operation
- Read Operation

Both operation modes consist of two bytes. The first byte contains the register address (7-bit) and bit[7] as MSB to indicate data direction. The second byte contains the data.

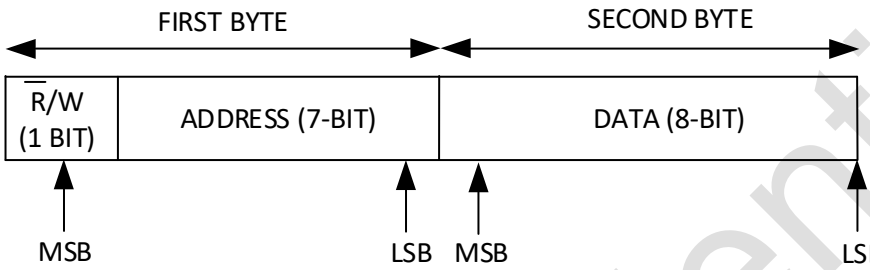


Figure 12. Transmission Protocol

6.4 Write Operation

The write operation consists of two bytes package. The first byte contains the 7-bit register address and a “1” in bit[7] indicates data write direction from the host to the chip. The second byte contains the data to be written into the specified register. The chip latches MOSI data on rising edges of the SCLK.

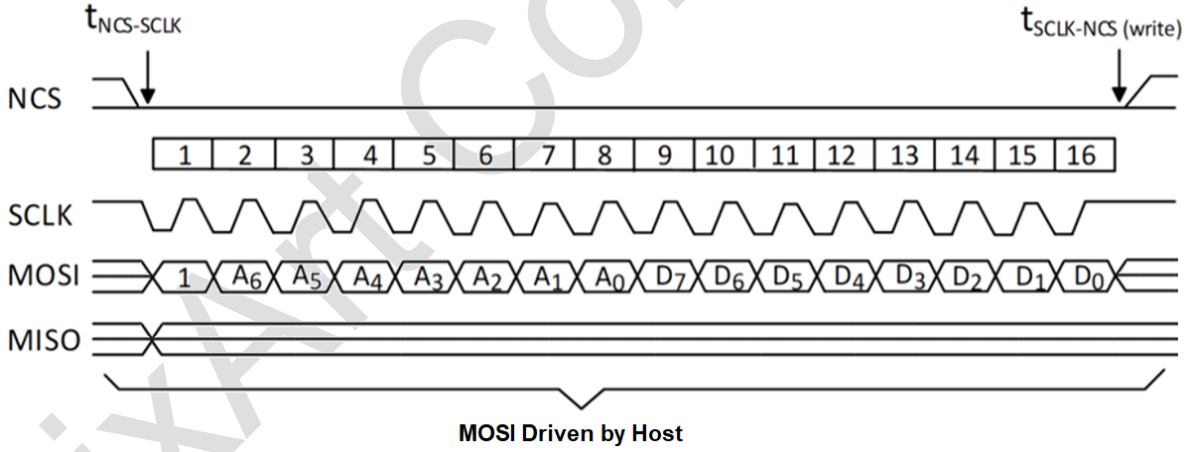


Figure 13. Write Operation

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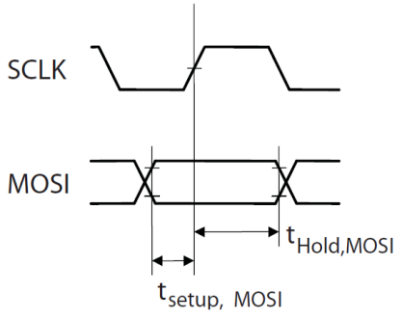


Figure 14. MOSI Set-up and Hold Time

6.5 Read Operation

The read operation is defined as data going from chip to the host, it is always initiated by the host and consists of two bytes. The first byte contains the 7-bit address, it is sent by the host over MOSI, and has a "0" on its MSB to indicate data direction. The second byte contains the data and it is driven by the chip over MISO. The chip outputs MISO bits on falling edges of SCLK and samples MOSI bits on every rising edge of SCLK.

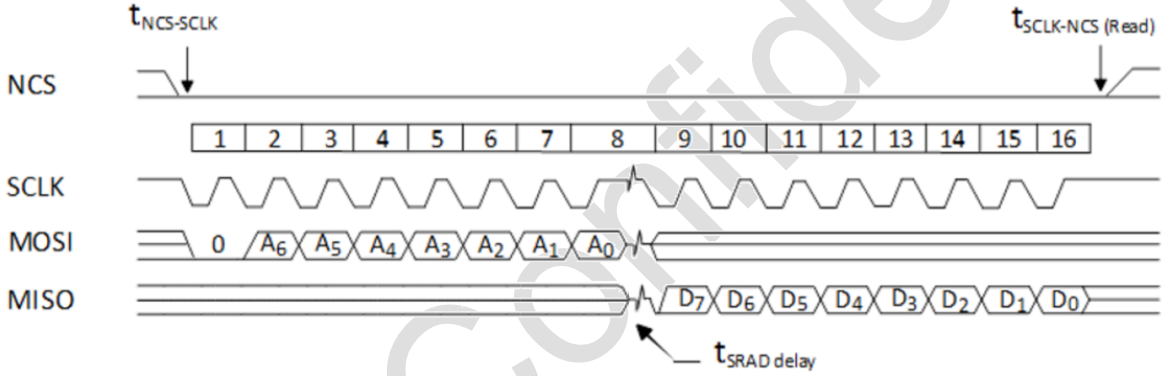


Figure 15. Read Operation

The minimum high state of SCLK is also the minimum MISO data hold time of the chip. Since the falling edge of SCLK is actually the start of the next read or write command, the chip will hold the state of data on MISO until the falling edge of SCLK.

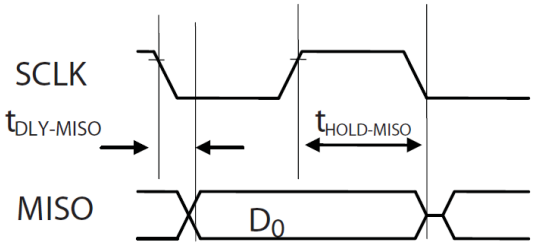


Figure 16. MISO Delay and Hold Time

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6.6 Burst Read

Burst read is a special serial port interface operation mode which can be used to reduce the serial transaction time for Motion Read. The speed improvement is achieved by continuous data clocking to read multiple data bytes with a single dedicated address and not requiring the delay period between data bytes. Refer to Section 7.4.4 for the detail.

NCS must be de-asserted to terminate burst mode after each burst read operation transaction is completed. However, the host can terminate the Burst Read operation as needed through de-assert the NCS.

After sending the register address, the host must wait for t_{SRAD}, and then begin reading data. All data bits can be read with no delay between bytes by driving SCLK at the normal rate. The data are latched into the output buffer after the last address bit is received. After the burst transmission is completed, the host NCS is de-assert for at least t_{BEXIT} to terminate burst read operation. The serial port is not available for use until assert of NCS.

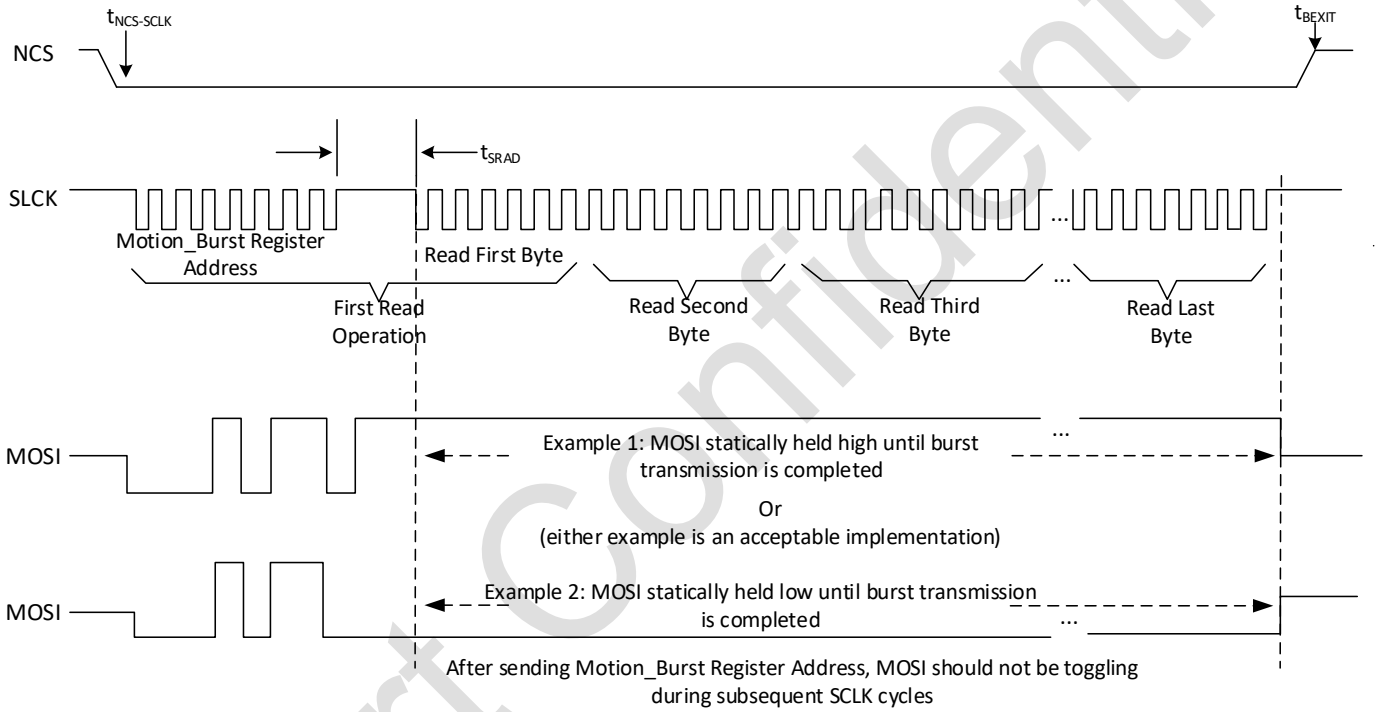


Figure 17. Burst Read Timing

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6.7 Required Timing between Read and Write Commands (t_{SXX})

There are minimum timing requirements between read and write commands on the serial port.

If the rising edge of the SCLK for the last data bit of the second write command occurs before the t_{SWW} delay, then the first write command may not complete correctly.

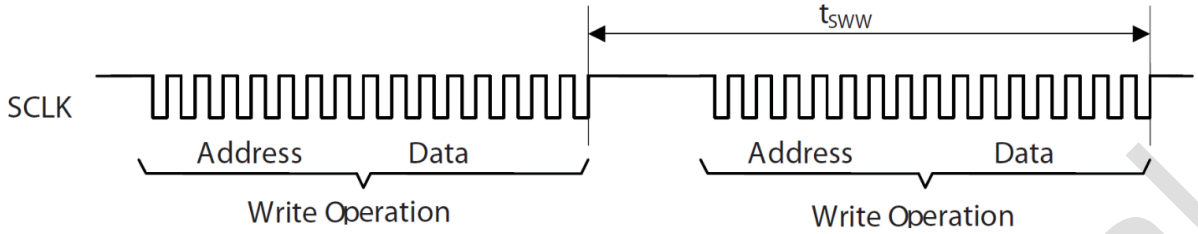


Figure 18. Timing between Two Write Commands

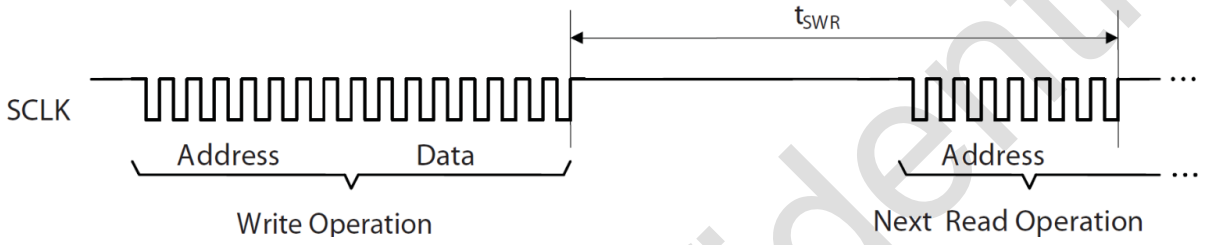


Figure 19. Timing between Write and Read Commands

If the rising edge of SCLK for the last address bit of the read command occurs before the t_{SWR} required delay, the write command may not complete correctly. During a read operation, SCLK must be delayed at least t_{SRAD} after the last address data bit to ensure that the chip has time to prepare the requested data.

The falling edge of SCLK for the first address bit of either the read or write command must be at least t_{SRR} or t_{SRW} after the last SCLK rising edge of the last data bit of the previous read operation. In addition, during a read operation SCLK must be delayed after the last address data bit to ensure that the chip has time to prepare the requested data.

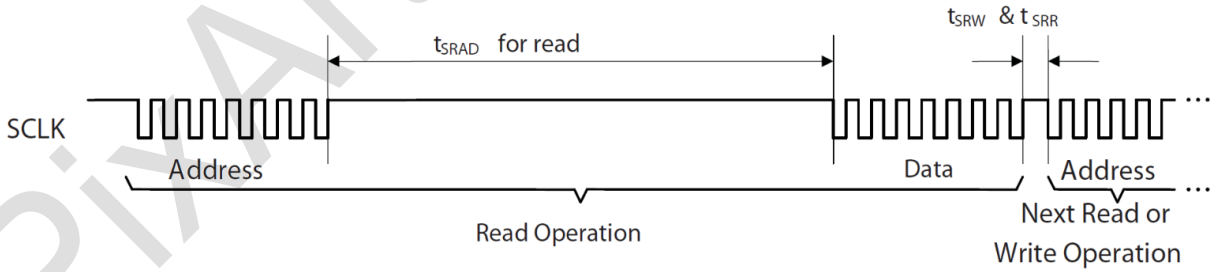


Figure 20. Timing between Read and Either Write or Subsequent Read Commands

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7.0 System Control

7.1 System Initialization

7.1.1 Initial Flow

The chip initialization sequence is as follows:

1. Apply power to VDD and VDDIO in any order, with less than 100ms of interval time in between VDD and VDDIO. Ensure all power supplies are stable.
2. Wait for at least 50 ms.
3. Drive NCS high for 1ms, and then low to reset the SPI port.
4. Write 0x5A to Register 0x3A (*Power_Up_Reset* register) or alternatively toggle the NRESET pin.
5. Wait for at least 5 ms.
6. Read *Product_ID* register to verify the PID before any other register read or write.
7. Refer to Section 7.1.2 to configure the initialization registers setting in order to achieve optimum performance.
8. Read registers 0x02, 0x03, 0x04, 0x05 and 0x06 to clear the motion data registers regardless of the motion pin state.

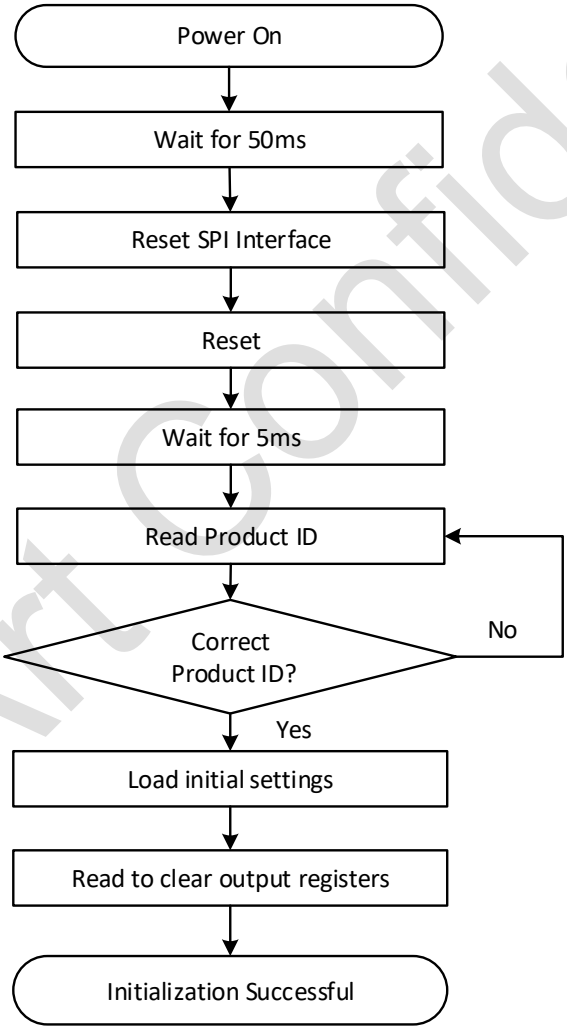


Figure 21. System Initialization Flow Chart

Note: If unable to read the correct Product ID, it is recommended to power off and power on or reset the chip again. If the problem persists, do check the SPI interface connection.

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7.1.2 Initialization Register Setting

Step in sequence as follows:

1. Write register 0x7F with value 0x07
2. Write register 0x40 with value 0x41
3. Write register 0x41 with value 0x0F
4. Write register 0x42 with value 0x28
5. Write register 0x43 with value 0x00
6. Write register 0x4A with value 0x08
7. Write register 0x51 with value 0x16
8. Write register 0x53 with value 0x04
9. Write register 0x7F with value 0x00
10. Wait for at least 1ms
11. Write register 0x7F with value 0x0E
12. Write register 0x55 with value 0xFC
13. Write register 0x56 with value 0XFF
14. Write register 0x7F with value 0x01
15. Write register 0x43 with value 0xF8
16. Write register 0x55 with value 0x28
17. Write register 0x57 with value 0x1A
18. Write register 0x5A with value 0x20
19. Write register 0x5B with value 0x0C
20. Write register 0x7F with value 0x05
21. Write register 0x46 with value 0x1B
22. Write register 0x5F with value 0x90
23. Write register 0x61 with value 0x31
24. Write register 0x6D with value 0xB8
25. Write register 0x6E with value 0xCF
26. Write register 0x7F with value 0x06
27. Write register 0x6B with value 0x20
28. Write register 0x6D with value 0x8F
29. Write register 0x44 with value 0x23
30. Write register 0x46 with value 0xB9
31. Write register 0x53 with value 0x03
32. Write register 0x61 with value 0x00
33. Write register 0x63 with value 0x00
34. Write register 0x7F with value 0x08
35. Write register 0x63 with value 0x08
36. Write register 0x6D with value 0x68
37. Write register 0x75with value 0x2C
38. Write register 0x79 with value 0xC8
39. Write register 0x7A with value 0x10
40. Write register 0x7C with value 0x34
41. Write register 0x7F with value 0x09
42. Write register 0x73 with value 0x0C
43. Write register 0x7F with value 0x0A
44. Write register 0x48 with value 0x80
45. Write register 0x50 with value 0x25
46. Write register 0x52 with value 0x40
47. Write register 0x53 with value 0x1F
48. Write register 0x56 with value 0x90
49. Write register 0x57 with value 0x01
50. Write register 0x4F with value 0xD0
51. Write register 0x59 with value 0x3F
52. Write register 0x7F with value 0x0C
53. Write register 0x4E with value 0x0A
54. Write register 0x48 with value 0x0A
55. Write register 0x42 with value 0x00
56. Write register 0x4C with value 0x4F
57. Write register 0x54 with value 0x1A
58. Write register 0x61 with value 0x18
59. Write register 0x62 with value 0x18
60. Write register 0x7F with value 0x0D
61. Write register 0x50 with value 0x66
62. Write register 0x5A with value 0x47
63. Write register 0x5B with value 0x66
64. Write register 0x5C with value 0xCB
65. Write register 0x5D with value 0x6C
66. Write register 0x71 with value 0xDC
67. Write register 0x70 with value 0x07
68. Write register 0x73 with value 0x16
69. Write register 0x72 with value 0x08
70. Write register 0x75 with value 0xDC
71. Write register 0x74 with value 0x07
72. Write register 0x77 with value 0x16
73. Write register 0x76 with value 0x08
74. Write register 0x7F with value 0x10
75. Write register 0x42 with value 0x1C
76. Write register 0x43 with value 0xD0
77. Write register 0x45 with value 0x11
78. Write register 0x46 with value 0x20

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79. Write register 0x48 with value 0x0A
80. Write register 0x49 with value 0x0F
81. Write register 0x4B with value 0x09
82. Write register 0x4C with value 0xC4
83. Write register 0x7F with value 0x14
84. Write register 0x43 with value 0x73
85. Write register 0x52 with value 0x19
86. Write register 0x62 with value 0x14
87. Write register 0x63 with value 0x04
88. Write register 0x65 with value 0x81
89. Write register 0x53 with value 0x50
90. Write register 0x54 with value 0xD8
91. Write register 0x7F with value 0x15
92. Write register 0x41 with value 0xA2
93. Write register 0x4C with value 0x04
94. Write register 0x56 with value 0x28
95. Write register 0x6F with value 0x28
96. Write register 0x70 with value 0x04
97. Write register 0x71 with value 0x04
98. Write register 0x77 with value 0x02
99. Write register 0x7D with value 0x10
100. Write register 0x45 with value 0xF4
101. Write register 0x47 with value 0x44
102. Write register 0x4B with value 0x12
103. Write register 0x7F with value 0x16
104. Write register 0x57 with value 0x43
105. Write register 0x58 with value 0x43
106. Write register 0x59 with value 0x43
107. Write register 0x5A with value 0x43
108. Write register 0x5D with value 0x32
109. Write register 0x5E with value 0x32
110. Write register 0x69 with value 0x1E
111. Write register 0x6A with value 0x22
112. Write register 0x6B with value 0x22
113. Write register 0x6C with value 0x22
114. Write register 0x6D with value 0x22
115. Write register 0x6E with value 0x18
116. Write register 0x6F with value 0x18
117. Write register 0x70 with value 0x18
118. Write register 0x71 with value 0x18
119. Write register 0x73 with value 0x00
120. Write register 0x74 with value 0xDA
121. Write register 0x75 with value 0xBE
122. Write register 0x78 with value 0x0E
123. Write register 0x7F with value 0x18
124. Write register 0x41 with value 0x02
125. Write register 0x43 with value 0x80
126. Write register 0x44 with value 0x1F
127. Write register 0x45 with value 0x01
128. Write register 0x46 with value 0x82
129. Write register 0x48 with value 0x65
130. Write register 0x4B with value 0x00
131. Write register 0x4C with value 0x2C
132. Write register 0x4D with value 0x38
133. Write register 0x4E with value 0x32
134. Write register 0x4F with value 0x7F
135. Write register 0x50 with value 0x34
136. Write register 0x51 with value 0x54
137. Write register 0x52 with value 0x38
138. Write register 0x54 with value 0x02
139. Write register 0x5B with value 0x1E
140. Write register 0x5C with value 0x60
141. Write register 0x5D with value 0x06
142. Write register 0x5E with value 0x02
143. Write register 0x5F with value 0x48
144. Write register 0x60 with value 0x44
145. Write register 0x61 with value 0x1C
146. Write register 0x62 with value 0x60
147. Write register 0x63 with value 0xFF
148. Write register 0x64 with value 0x00
149. Write register 0x65 with value 0x75
150. Write register 0x67 with value 0x4C
151. Write register 0x69 with value 0x85
152. Write register 0x6B with value 0x60
153. Write register 0x6D with value 0xF5
154. Write register 0x6E with value 0x06
155. Write register 0x6F with value 0x06
156. Write register 0x70 with value 0x22
157. Write register 0x71 with value 0xF2
158. Write register 0x72 with value 0xF6
159. Write register 0x76 with value 0x00
160. Write register 0x77 with value 0x32
161. Write register 0x78 with value 0x88
162. Write register 0x79 with value 0xCC

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163. Write register 0x7A with value 0x00
164. Write register 0x7E with value 0xB0
165. Write register 0x7F with value 0x19
166. Write register 0x40 with value 0x43
167. Write register 0x41 with value 0x32
168. Write register 0x44 with value 0x32
169. Write register 0x46 with value 0xE6
170. Write register 0x4B with value 0x01
171. Write register 0x4C with value 0x01
172. Write register 0x52 with value 0x15
173. Write register 0x53 with value 0x12
174. Write register 0x54 with value 0x23
175. Write register 0x56 with value 0x78
176. Write register 0x5F with value 0x7F
177. Write register 0x6E with value 0x8F
178. Write register 0x7F with value 0x00
179. Write register 0x5B with value 0x40
180. Write register 0x47 with value 0x01
181. Write register 0x57 with value 0x20
182. Write register 0x64 with value 0x08
183. Write register 0x65 with value 0x07
184. Write register 0x7F with value 0x07
185. Write register 0x40 with value 0x40
186. Write register 0x7F with value 0x00
187. Write register 0x40 with value 0x80
188. Write register 0x55 with value 0x02
189. Write register 0x23 with value 0x70
190. Write register 0x22 with value 0x01
191. Wait for at least 1ms
192. Read register 0x6C at 1ms interval until value 0x80 is obtained or read up to 45 times, this register read interval must be carried out at 1ms interval with timing tolerance of ±1%.
193. Wait for minimum 5ms
194. Write register 0x22 with value 0x00
195. Write register 0x55 with value 0x00
196. Write register 0x40 with value 0x00
197. Write register 0x68 with value 0x01

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7.2 Operation Mode

With the enhanced navigation technology, the Glass mode is incorporated into both Gaming mode and Office mode. The chip automatically switches between Gaming/Office mode and Glass mode when tracking on different surfaces. There are 4 operation mode settings that can be configured into the chip by writing a set of register commands, refer to section 7.2.1 for the details.

Operation Mode Setting:

- High Performance Gaming and Glass
- Low Power Gaming and Glass
- Corded Gaming and Glass
- Office and Office Glass

During the operation, the host is responsible for changing the polling rate and the REST state when the chip is operating in different operation mode. The maximum polling rate and the recommended REST setting are shown in the following table.

MOTION register bit[2] (*OP_Mode*) can be used to detect the chip's operation mode. During the operation, read *OP_Mode* to change the polling rate and REST setting.

For example, if the chip is operating in Corded Gaming and Glass Mode:

- When *OP_Mode* equals to "1" (Corded Gaming Mode), set to 8 kHz polling rate and disable REST
- When *OP_Mode* equals to "0" (Glass Mode), set to 1 kHz polling rate and enable REST

Table 16. Maximum Polling Rate and Recommended Rest State

Operation Mode		Max. Polling Rate	Recommended REST State	*OP_Mode*
Gaming Mode	High Performance	1kHz	REST Enabled	1
	Low Power	1kHz	REST Enabled	1
	Corded	8kHz	REST Disabled	1
	Glass	1kHz	REST Enabled	0
Office Mode	Office	125Hz	REST Enabled	1
	Office Glass	125Hz		0

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7.2.1 Operation Mode Setting

There are 4 operation mode settings can be programmed through the register commands. Upon the chip power-on as per the recommended initial flow in Section 7.1.1, the chip is configured to High Performance Gaming and Glass Mode by default. It can be configured to other setting without the need of re-initialization.

7.2.2 High Performance Gaming and Glass Mode (Default)

1. Write register 0x7F with value 0x07
2. Write register 0x40 with value 0x41
3. Write register 0x7F with value 0x05
4. Write register 0x43 with value 0xE4
5. Write register 0x44 with value 0x05
6. Write register 0x49 with value 0x10
7. Write register 0x51 with value 0x40
8. Write register 0x53 with value 0x40
9. Write register 0x55 with value 0xFB
10. Write register 0x5B with value 0xFB
11. Write register 0x5F with value 0x90
12. Write register 0x61 with value 0x31
13. Write register 0x6D with value 0xB8
14. Write register 0x6E with value 0xCF
15. Write register 0x7B with value 0x10
16. Write register 0x7F with value 0x06
17. Write register 0x53 with value 0x03
18. Write register 0x62 with value 0x01
19. Write register 0x7A with value 0x02
20. Write register 0x6B with value 0x20
21. Write register 0x6D with value 0x8F
22. Write register 0x6E with value 0x70
23. Write register 0x6F with value 0x04
24. Write register 0x7F with value 0x07
25. Write register 0x42 with value 0x28
26. Write register 0x43 with value 0x00
27. Write register 0x7F with value 0x09
28. Write register 0x40 with value 0x01
29. Write register 0x43 with value 0x13
30. Write register 0x44 with value 0x88
31. Write register 0x47 with value 0x00
32. Write register 0x4F with value 0x0C
33. Write register 0x51 with value 0x04
34. Write register 0x55 with value 0x3F
35. Write register 0x56 with value 0x3F
36. Write register 0x59 with value 0x0F
37. Write register 0x5A with value 0x0F
38. Write register 0x73 with value 0x0c
39. Write register 0x7F with value 0x0A
40. Write register 0x4A with value 0x14
41. Write register 0x7F with value 0x19
42. Write register 0x41 with value 0x32
43. Write register 0x47 with value 0x12
44. Write register 0x4B with value 0x01
45. Write register 0x4C with value 0x01
46. Write register 0x7F with value 0x0C
47. Write register 0x4a with value 0x1c
48. Write register 0x4b with value 0x14
49. Write register 0x4c with value 0x4F
50. Write register 0x4d with value 0x02
51. Write register 0x50 with value 0x00
52. Write register 0x51 with value 0x01
53. Write register 0x53 with value 0x16
54. Write register 0x55 with value 0x10
55. Write register 0x62 with value 0x18
56. Write register 0x7F with value 0x14
57. Write register 0x62 with value 0x14
58. Write register 0x7F with value 0x18
59. Write register 0x48 with value 0x65
60. Write register 0x50 with value 0x34
61. Write register 0x51 with value 0x54
62. Write register 0x52 with value 0x38
63. Write register 0x53 with value 0x5F
64. Write register 0x55 with value 0x7F
65. Write register 0x61 with value 0x1C
66. Write register 0x62 with value 0x60
67. Write register 0x63 with value 0xFF
68. Write register 0x64 with value 0x00
69. Write register 0x70 with value 0x22
70. Write register 0x71 with value 0xF2

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71. Write register 0x72 with value 0xF6
72. Write register 0x77 with value 0x32
73. Write register 0x79 with value 0xCC
74. Write register 0x4c with value 0x2C
75. Write register 0x4d with value 0x38
76. Write register 0x4e with value 0x32
77. Write register 0x5a with value 0x30
78. Write register 0x5b with value 0x1E
79. Write register 0x5d with value 0x06
80. Write register 0x5e with value 0x02
81. Write register 0x6d with value 0xF5
82. Write register 0x6e with value 0x06
83. Write register 0x6f with value 0x06
84. Write register 0x7F with value 0x00
85. Write register 0x4F with value 0x4F
86. Write register 0x4E with value 0x00
87. Write register 0x52 with value 0x4F
88. Write register 0x51 with value 0x00
89. Write register 0x47 with value 0x01
90. Write register 0x54 with value 0x53
91. Write register 0x5A with value 0x80
92. Write register 0x78 with value 0x01
93. Write register 0x79 with value 0x9C
94. Write register 0x40 bit[1:0] with value “0”
95. Write register 0x7F with value 0x07
96. Write register 0x40 with value 0x40
97. Write register 0x7F with value 0x00

Note: For configuring register 0x40, avoid overwriting other bit within the register.

7.2.3 Low Power Gaming and Glass Mode

1. Write register 0x7F with value 0x07
2. Write register 0x40 with value 0x41
3. Write register 0x7F with value 0x05
4. Write register 0x43 with value 0xE4
5. Write register 0x44 with value 0x05
6. Write register 0x49 with value 0x10
7. Write register 0x51 with value 0x40
8. Write register 0x53 with value 0x40
9. Write register 0x55 with value 0xFC
10. Write register 0x5B with value 0xFC
11. Write register 0x5F with value 0x90
12. Write register 0x61 with value 0x3B
13. Write register 0x6D with value 0xB8
14. Write register 0x6E with value 0xDF
15. Write register 0x7B with value 0x10
16. Write register 0x7F with value 0x06
17. Write register 0x53 with value 0x03
18. Write register 0x62 with value 0x01
19. Write register 0x7A with value 0x02
20. Write register 0x6B with value 0x20
21. Write register 0x6D with value 0x8F
22. Write register 0x6E with value 0x70
23. Write register 0x6F with value 0x04
24. Write register 0x7F with value 0x07
25. Write register 0x42 with value 0x28
26. Write register 0x43 with value 0x00
27. Write register 0x7F with value 0x09
28. Write register 0x40 with value 0x01
29. Write register 0x43 with value 0x13
30. Write register 0x44 with value 0x88
31. Write register 0x47 with value 0x00
32. Write register 0x4F with value 0x0C
33. Write register 0x51 with value 0x04
34. Write register 0x55 with value 0x3F
35. Write register 0x56 with value 0x3F
36. Write register 0x59 with value 0x0F
37. Write register 0x5A with value 0x0F
38. Write register 0x73 with value 0x0c
39. Write register 0x7F with value 0x0A
40. Write register 0x4A with value 0x11
41. Write register 0x7F with value 0x19
42. Write register 0x41 with value 0x32
43. Write register 0x47 with value 0x12
44. Write register 0x4B with value 0x01
45. Write register 0x4C with value 0x01
46. Write register 0x7F with value 0x0C
47. Write register 0x4a with value 0x1c
48. Write register 0x4b with value 0x14

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49. Write register 0x4c with value 0x4F
50. Write register 0x4d with value 0x02
51. Write register 0x50 with value 0x00
52. Write register 0x51 with value 0x01
53. Write register 0x53 with value 0x16
54. Write register 0x55 with value 0x10
55. Write register 0x62 with value 0x18
56. Write register 0x7F with value 0x14
57. Write register 0x62 with value 0x14
58. Write register 0x7F with value 0x18
59. Write register 0x48 with value 0x65
60. Write register 0x50 with value 0x34
61. Write register 0x51 with value 0x54
62. Write register 0x52 with value 0x38
63. Write register 0x53 with value 0x5F
64. Write register 0x55 with value 0x7F
65. Write register 0x61 with value 0x1C
66. Write register 0x62 with value 0x60
67. Write register 0x63 with value 0xFF
68. Write register 0x64 with value 0x00
69. Write register 0x70 with value 0x22
70. Write register 0x71 with value 0xF2
71. Write register 0x72 with value 0xF6
72. Write register 0x77 with value 0x32
73. Write register 0x79 with value 0xCC
74. Write register 0x4c with value 0x2C
75. Write register 0x4d with value 0x38
76. Write register 0x4e with value 0x32
77. Write register 0x5a with value 0x30
78. Write register 0x5b with value 0x1E
79. Write register 0x5d with value 0x06
80. Write register 0x5e with value 0x02
81. Write register 0x6d with value 0xF5
82. Write register 0x6e with value 0x06
83. Write register 0x6f with value 0x06
84. Write register 0x7F with value 0x00
85. Write register 0x4F with value 0x4F
86. Write register 0x4E with value 0x00
87. Write register 0x52 with value 0x4F
88. Write register 0x51 with value 0x00
89. Write register 0x47 with value 0x01
90. Write register 0x54 with value 0x53
91. Write register 0x5A with value 0x80
92. Write register 0x78 with value 0x01
93. Write register 0x79 with value 0x9C
94. Write register 0x40 bit[1:0] with value “01”
95. Write register 0x7F with value 0x07
96. Write register 0x40 with value 0x40
97. Write register 0x7F with value 0x00

Note: For configuring register 0x40, avoid overwriting other bit within the register.

7.2.4 Corded Gaming and Glass Mode (< 7000cpi)

When selecting a resolution of lower than 7000cpi, write the following register settings to configure the chip to Corded Gaming and Glass Mode.

1. Write register 0x7F with value 0x07
2. Write register 0x40 with value 0x41
3. Write register 0x7F with value 0x05
4. Write register 0x43 with value 0x64
5. Write register 0x44 with value 0x05
6. Write register 0x49 with value 0x20
7. Write register 0x51 with value 0x40
8. Write register 0x53 with value 0x40
9. Write register 0x55 with value 0xFB
10. Write register 0x5B with value 0xFB
11. Write register 0x5F with value 0x90
12. Write register 0x61 with value 0x31
13. Write register 0x6D with value 0xB8
14. Write register 0x6E with value 0xCF
15. Write register 0x7B with value 0x50
16. Write register 0x7F with value 0x06
17. Write register 0x53 with value 0x03
18. Write register 0x62 with value 0x02
19. Write register 0x7A with value 0x03
20. Write register 0x6B with value 0x20
21. Write register 0x6D with value 0x8F
22. Write register 0x6E with value 0x70

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23. Write register 0x6F with value 0x07
24. Write register 0x7F with value 0x07
25. Write register 0x42 with value 0x14
26. Write register 0x43 with value 0x00
27. Write register 0x7F with value 0x09
28. Write register 0x40 with value 0x01
29. Write register 0x43 with value 0x23
30. Write register 0x44 with value 0x88
31. Write register 0x47 with value 0x00
32. Write register 0x4F with value 0x0C
33. Write register 0x51 with value 0x04
34. Write register 0x55 with value 0x3F
35. Write register 0x56 with value 0x3F
36. Write register 0x59 with value 0x0F
37. Write register 0x5A with value 0x0F
38. Write register 0x73 with value 0x0c
39. Write register 0x7F with value 0x0A
40. Write register 0x4A with value 0x14
41. Write register 0x7F with value 0x19
42. Write register 0x41 with value 0x32
43. Write register 0x47 with value 0x24
44. Write register 0x4B with value 0x02
45. Write register 0x4C with value 0x02
46. Write register 0x7F with value 0x0C
47. Write register 0x4a with value 0x20
48. Write register 0x4b with value 0x1f
49. Write register 0x4c with value 0x5c
50. Write register 0x4d with value 0x90
51. Write register 0x50 with value 0x14
52. Write register 0x51 with value 0x15
53. Write register 0x53 with value 0x1e
54. Write register 0x55 with value 0x02
55. Write register 0x62 with value 0x02
56. Write register 0x7F with value 0x14
57. Write register 0x62 with value 0x14
58. Write register 0x7F with value 0x18
59. Write register 0x48 with value 0x55
60. Write register 0x50 with value 0x18
61. Write register 0x51 with value 0x40
62. Write register 0x52 with value 0x20
63. Write register 0x53 with value 0x38
64. Write register 0x55 with value 0x68
65. Write register 0x61 with value 0x0a
66. Write register 0x62 with value 0x1a
67. Write register 0x63 with value 0x48
68. Write register 0x64 with value 0x40
69. Write register 0x70 with value 0x22
70. Write register 0x71 with value 0x88
71. Write register 0x72 with value 0x88
72. Write register 0x77 with value 0x22
73. Write register 0x79 with value 0xcb
74. Write register 0x4c with value 0x20
75. Write register 0x4d with value 0x30
76. Write register 0x4e with value 0x28
77. Write register 0x5a with value 0x30
78. Write register 0x5b with value 0x24
79. Write register 0x5d with value 0x02
80. Write register 0x5e with value 0x01
81. Write register 0x6d with value 0x85
82. Write register 0x6e with value 0x0c
83. Write register 0x6f with value 0x0a
84. Write register 0x7F with value 0x00
85. Write register 0x4F with value 0x4F
86. Write register 0x4E with value 0x00
87. Write register 0x52 with value 0x4F
88. Write register 0x51 with value 0x00
89. Write register 0x47 with value 0x01
90. Write register 0x54 with value 0x55
91. Write register 0x5A with value 0x80
92. Write register 0x40 with value 0x83
93. Write register 0x7F with value 0x07
94. Write register 0x40 with value 0x40
95. Write register 0x7F with value 0x00

Note: For configuring register 0x40, avoid overwriting other bit within the register.

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7.2.5 Corded Gaming and Glass Mode (≥ 7000cpi)

When selecting a resolution of 7000cpi or higher, write the following register settings to configure the chip to Corded Gaming and Glass Mode.

1. Write register 0x7F with value 0x07
2. Write register 0x40 with value 0x41
3. Write register 0x7F with value 0x05
4. Write register 0x43 with value 0x64
5. Write register 0x44 with value 0x05
6. Write register 0x49 with value 0x20
7. Write register 0x51 with value 0x10
8. Write register 0x53 with value 0x40
9. Write register 0x55 with value 0xFB
10. Write register 0x5B with value 0xFB
11. Write register 0x5F with value 0x80
12. Write register 0x61 with value 0x31
13. Write register 0x6D with value 0xA7
14. Write register 0x6E with value 0xCD
15. Write register 0x7B with value 0x50
16. Write register 0x7F with value 0x06
17. Write register 0x53 with value 0x02
18. Write register 0x62 with value 0x02
19. Write register 0x7A with value 0x03
20. Write register 0x6B with value 0x20
21. Write register 0x6D with value 0x98
22. Write register 0x6E with value 0x80
23. Write register 0x6F with value 0x07
24. Write register 0x7F with value 0x07
25. Write register 0x42 with value 0x14
26. Write register 0x43 with value 0x00
27. Write register 0x7F with value 0x09
28. Write register 0x40 with value 0x03
29. Write register 0x43 with value 0x23
30. Write register 0x44 with value 0x98
31. Write register 0x47 with value 0x18
32. Write register 0x4F with value 0x00
33. Write register 0x51 with value 0x11
34. Write register 0x55 with value 0x54
35. Write register 0x56 with value 0x54
36. Write register 0x59 with value 0x20
37. Write register 0x5A with value 0x20
38. Write register 0x73 with value 0x0e
39. Write register 0x7F with value 0x0A
40. Write register 0x4A with value 0x14
41. Write register 0x7F with value 0x19
42. Write register 0x41 with value 0x32
43. Write register 0x47 with value 0x24
44. Write register 0x4B with value 0x02
45. Write register 0x4C with value 0x02
46. Write register 0x7F with value 0x0C
47. Write register 0x4a with value 0x20
48. Write register 0x4b with value 0x1f
49. Write register 0x4c with value 0x5c
50. Write register 0x4d with value 0x90
51. Write register 0x50 with value 0x14
52. Write register 0x51 with value 0x15
53. Write register 0x53 with value 0x1e
54. Write register 0x55 with value 0x02
55. Write register 0x62 with value 0x02
56. Write register 0x7F with value 0x14
57. Write register 0x62 with value 0x14
58. Write register 0x7F with value 0x18
59. Write register 0x48 with value 0x55
60. Write register 0x50 with value 0x18
61. Write register 0x51 with value 0x40
62. Write register 0x52 with value 0x20
63. Write register 0x53 with value 0x38
64. Write register 0x55 with value 0x68
65. Write register 0x61 with value 0x0a
66. Write register 0x62 with value 0x1a
67. Write register 0x63 with value 0x48
68. Write register 0x64 with value 0x40
69. Write register 0x70 with value 0x22
70. Write register 0x71 with value 0x88
71. Write register 0x72 with value 0x88
72. Write register 0x77 with value 0x22
73. Write register 0x79 with value 0xcb
74. Write register 0x4c with value 0x20
75. Write register 0x4d with value 0x30
76. Write register 0x4e with value 0x28

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77. Write register 0x5a with value 0x30
78. Write register 0x5b with value 0x24
79. Write register 0x5d with value 0x02
80. Write register 0x5e with value 0x01
81. Write register 0x6d with value 0x85
82. Write register 0x6e with value 0x0c
83. Write register 0x6f with value 0x0a
84. Write register 0x7F with value 0x00
85. Write register 0x4F with value 0x95
86. Write register 0x4E with value 0x00
87. Write register 0x52 with value 0x95
88. Write register 0x51 with value 0x00
89. Write register 0x47 with value 0x01
90. Write register 0x54 with value 0x55
91. Write register 0x5A with value 0x80
92. Write register 0x40 with value 0x83
93. Write register 0x7F with value 0x07
94. Write register 0x40 with value 0x40
95. Write register 0x7F with value 0x00

Note: For configuring register 0x40, avoid overwriting other bit within the register.

7.2.6 Office and Office Glass Mode

1. Write register 0x7F with value 0x07
2. Write register 0x40 with value 0x41
3. Write register 0x7F with value 0x05
4. Write register 0x43 with value 0xE4
5. Write register 0x44 with value 0x05
6. Write register 0x49 with value 0x10
7. Write register 0x51 with value 0x28
8. Write register 0x53 with value 0x30
9. Write register 0x55 with value 0xFB
10. Write register 0x5B with value 0xFB
11. Write register 0x5F with value 0x90
12. Write register 0x61 with value 0x3B
13. Write register 0x6D with value 0xB8
14. Write register 0x6E with value 0xDF
15. Write register 0x7B with value 0x10
16. Write register 0x7F with value 0x06
17. Write register 0x53 with value 0x03
18. Write register 0x62 with value 0x01
19. Write register 0x7A with value 0x02
20. Write register 0x6B with value 0x20
21. Write register 0x6D with value 0x8F
22. Write register 0x6E with value 0x70
23. Write register 0x6F with value 0x04
24. Write register 0x7F with value 0x07
25. Write register 0x42 with value 0x28
26. Write register 0x43 with value 0x00
27. Write register 0x7F with value 0x09
28. Write register 0x40 with value 0x01
29. Write register 0x43 with value 0x13
30. Write register 0x44 with value 0x88
31. Write register 0x47 with value 0x00
32. Write register 0x4F with value 0x0C
33. Write register 0x51 with value 0x04
34. Write register 0x55 with value 0x3F
35. Write register 0x56 with value 0x3F
36. Write register 0x59 with value 0x0F
37. Write register 0x5A with value 0x0F
38. Write register 0x73 with value 0x0c
39. Write register 0x7F with value 0x0A
40. Write register 0x4A with value 0x14
41. Write register 0x7F with value 0x19
42. Write register 0x41 with value 0x53
43. Write register 0x47 with value 0x64
44. Write register 0x4B with value 0x02
45. Write register 0x4C with value 0x02
46. Write register 0x7F with value 0x0C
47. Write register 0x4a with value 0x1c
48. Write register 0x4b with value 0x14
49. Write register 0x4c with value 0x4F
50. Write register 0x4d with value 0x02
51. Write register 0x50 with value 0x00
52. Write register 0x51 with value 0x01
53. Write register 0x53 with value 0x16
54. Write register 0x55 with value 0x10
55. Write register 0x62 with value 0x18
56. Write register 0x7F with value 0x14

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57. Write register 0x62 with value 0x14
58. Write register 0x7F with value 0x18
59. Write register 0x48 with value 0x65
60. Write register 0x50 with value 0x34
61. Write register 0x51 with value 0x54
62. Write register 0x52 with value 0x38
63. Write register 0x53 with value 0x5F
64. Write register 0x55 with value 0x7F
65. Write register 0x61 with value 0x1C
66. Write register 0x62 with value 0x60
67. Write register 0x63 with value 0xFF
68. Write register 0x64 with value 0x00
69. Write register 0x70 with value 0x22
70. Write register 0x71 with value 0xF2
71. Write register 0x72 with value 0xF6
72. Write register 0x77 with value 0x32
73. Write register 0x79 with value 0xCC
74. Write register 0x4c with value 0x2C
75. Write register 0x4d with value 0x38
76. Write register 0x4e with value 0x32
77. Write register 0x5a with value 0x30
78. Write register 0x5b with value 0x1E
79. Write register 0x5d with value 0x06
80. Write register 0x5e with value 0x02
81. Write register 0x6d with value 0xF5
82. Write register 0x6e with value 0x06
83. Write register 0x6f with value 0x06
84. Write register 0x7F with value 0x00
85. Write register 0x4F with value 0x4F
86. Write register 0x4E with value 0x00
87. Write register 0x52 with value 0x4F
88. Write register 0x51 with value 0x00
89. Write register 0x47 with value 0x01
90. Write register 0x54 with value 0x52
91. Write register 0x5A with value 0x80
92. Write register 0x78 with value 0x0A
93. Write register 0x79 with value 0x10
94. Write register 0x40 bit[1:0] with value “10”
95. Write register 0x7F with value 0x07
96. Write register 0x40 with value 0x40
97. Write register 0x7F with value 0x00

Note: For configuring register 0x40, avoid overwriting other bit within the register.

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7.3 Registers Access

The host can access register through SPI interface. Refer to section 6.0 for detail.

7.3.1 Registers Address Mapping

The chip has two types of register assignment. There are register bank and register address mapping. Each register bank consists of 7 bits width address mapping (0x00 to 0x7F).

7.3.2 Register Bank Switching

Write register 0x7F to switch the register bank. Upon the chip initialization, the chip is set to bank 0x00 as default. It is highly recommended to switch back to register bank 0x00 after configure a register setting from other register bank.

For example, to write *Lift_Config* register with value of 0x85 (Bank: 0x18, Address: 0x7A):

1. Write register 0x7F with value 0x18
2. Write register 0x7A with value 0x85
3. Write register 0x7F with value 0x00

7.3.3 Reserved Bit Protection

The chip supports only byte length write operation. When the write operation is performed to the register with reserved bit, it is required to protect the reserved bit from being overwritten.

To protect the reserved bit in a register byte, the sequence as follow:

1. Read and store the present register value
2. Perform bit operation towards the read back value in order to overwrite the specific bit
3. Write the new value to the register

Example:

To disable the Rest State in Register 0x40 (set bit[7:1])

Read register 0x40 and store in VarA
VarA |= 0x80
Write register 0x40 with value VarA

To enable the Rest State in Register 0x40 (set bit[7:0])

Read register 0x40 and store in VarA
VarA &= ~ 0x80
Write register 0x40 with value VarA

7.3.4 Related Register

Usage	Register Name	Bank	Address
Bank Switching	Bank_Switch	NA	0x7F

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7.4 Output

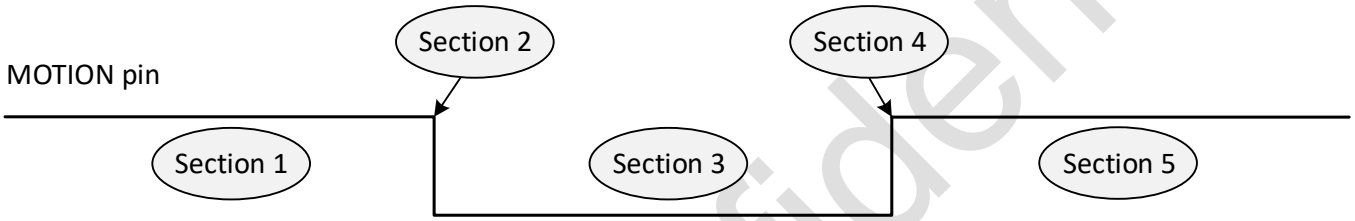
7.4.1 Motion Data

When the chip detects motion, the Motion bit (*MOTION* register bit [7]) will be set and MOTION pin will be asserted. The motion data (X-movement and Y-movement) is accumulated and stored in chip's internal register *Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H*.

7.4.2 Motion Bit and Motion Pin Interrupt

The motion bit in the *Motion* register (address 0x02) is an indicator flag for motion detection. Prior reading motion data registers (Delta X and Delta Y registers), the host must check motion bit in the *Motion* register (address 0x02). Motion bit will be updated as 1 when the motion data are available.

The motion pin is an interrupt (active low) pin that triggers the host when motion has occurred. The MOTION pin asserts whenever the motion bit in *Motion* register (address 0x02) is "1".



Section 1 : No motion detected, motion data are invalid. Motion bit= 0 and motion pin de-asserted.
Section 2 : When motion detected, motion pin is asserted, motion bit= 1 and motion data accumulation begins.
Section 3 : Motion data is available and wait for host to access.
Section 4 : Upon motion data register is accessed, motion bit= 0, MOTION pin is de-asserted and motion data registers are automatic cleared.
Section 5 : No motion detected, motion data are invalid. Motion bit=0 and motion pin de-asserted.

Figure 22. Motion Interrupt Pin Function

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7.4.3 Output Access

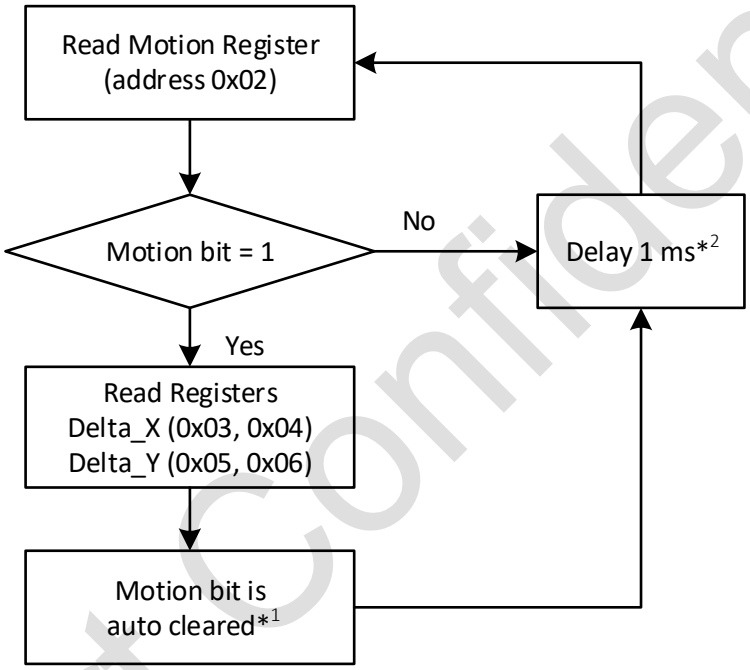
The accumulated motion data are stored in *Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H* registers. The data can be accessed through single or burst read operation.

There are two methods to read out the motion data:

1. Polling method
2. Interrupt method

- Polling Method

By reading and checking the register *Motion_Status* (address 0x02) periodically, the host controller can get the motion data through SPI interface. The polling interval can be varied where it depends on the host capability and requirement of the applications.

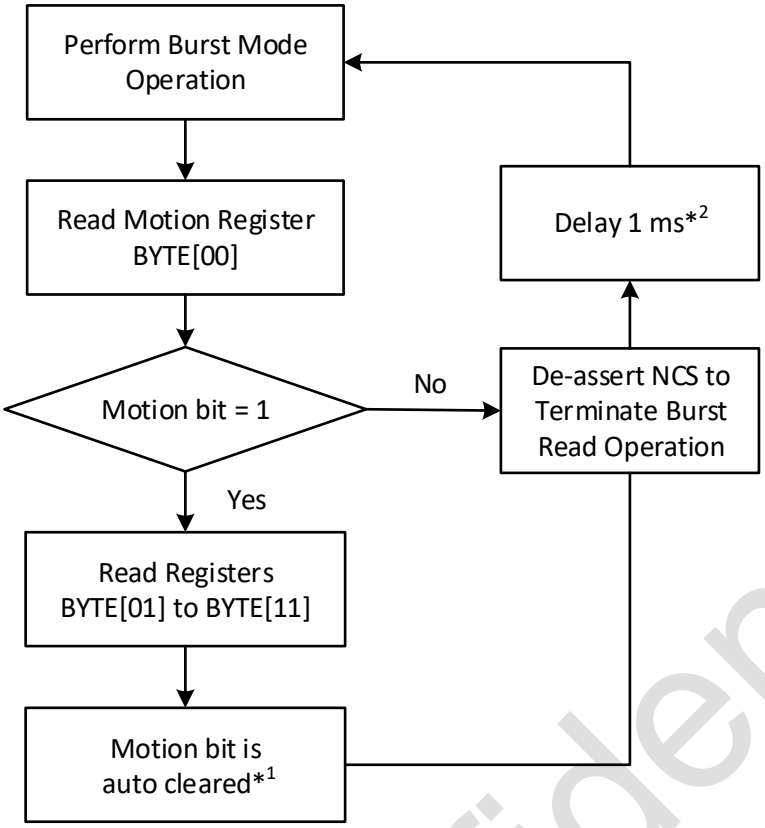


Note:
*1: Motion bit is auto cleared after reading *MOTION* data register
*2: The polling interval is configured by the host

Figure 23. Polling Method – Single Read

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Note:
*1: Motion bit is auto cleared after reading *MOTION* register
*2: The polling interval is configured by the host

Figure 24. Polling Method – Burst Read

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- Interrupt Method

When the chip detects the occurrence of motion, the MOTION pin is asserted and triggers the host to read data.

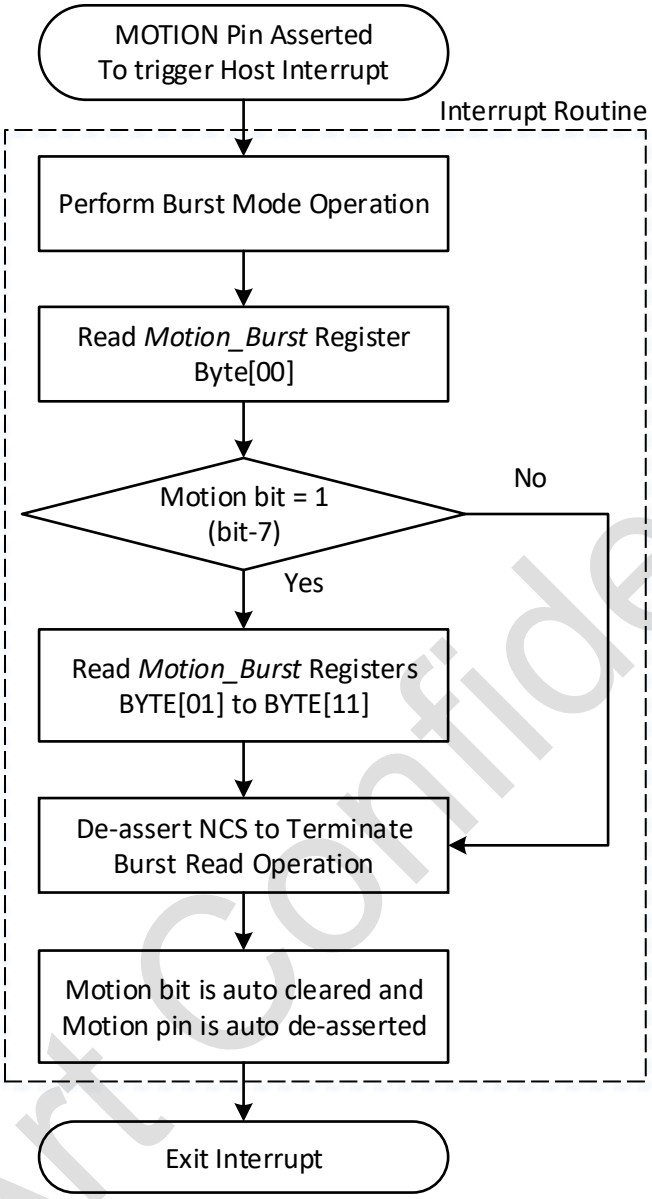


Figure 25. Interrupt Method – Burst Read

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7.4.4 Burst Read Operation

Burst read operation is a special serial port interface operation mode which use to minimize the serial transaction time for accessing motion data. The speed improvement is achieved by continuous data clocking from the host for multiple registers read without further specify the register address. In the burst read operation, NCS must be remained asserted until burst mode transaction is fully completed.

Note: A single read of any motion related registers (0x02 to 0x06) must be avoided during burst mode.

Read the *Motion_Burst* register to activate burst read. The chip will output the motion data with the following motion burst report order.

Motion burst report:

BYTE[00]= Motion
BYTE[01]= Observation
BYTE[02]= Delta_X_L
BYTE[03]= Delta_X_H
BYTE[04]= Delta_Y_L
BYTE[05]= Delta_Y_H
BYTE[06]= SQUAL
BYTE[07]= RawData_Sum
BYTE[08]= Maximum_RawData
BYTE[09]= Minimum_RawData
BYTE[10]= Shutter_Upper
BYTE[11]= Shutter_Lower

Procedure to start motion burst:

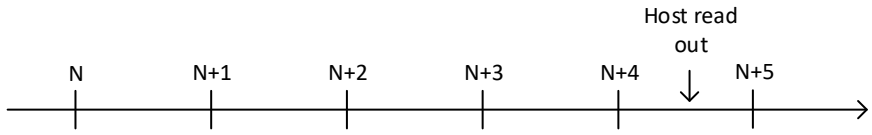
1. Assert NCS signal, and wait for $t_{NCS\text{-}SCLK}$ delay.
2. Send *Motion_Burst* address (0x16). After sending this address, MOSI can be held either high or low until the burst transmission is completed (MOSI must not be toggled during subsequent SCLK cycles).
3. Wait for t_{SRAD}.
4. Start reading SPI Data continuously up to 12 bytes.
5. Motion burst can be terminated by pulling the NCS high for at least t_{BEXI} as needed.
6. To read next motion burst data, repeat from step 1.

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7.4.5 Data Lost and Corruption

Motion data register storage limit is -32768 to 32767 for X-axis and Y-axis. When motion is detected, motion data will be generated, accumulated, and stored in registers. If host didn't read out motion data before data reach the limit, the data corruption may occur.



	The change in X-axis motion data by frame	0	+10	+10	-10	-10	+10
Condition1: (no data lost)	X-axis motion data in internal buffer	0	10	20	10	0	10
Condition2: (data lost)	X-axis motion data in internal buffer	32750	32760	**32767**	32757	32747	10

Figure 26. Example of Data Lost When Internal Buffer Reach limit

From frame N to frame N+4, the total change of motion data is 0.

- Condition1

  If motion data register value is 0 in frame N, the host reads motion data value 0 in between frames N+4 and N+5.

- Condition2

  If motion data register value is 32750 in frame N, the host reads motion data value 32747 in between frames N+4 and N+5.

The above two condition movement behavior are identical. However, condition 2 register hitting limit of 32767, host read value 32747 instead of 32750 where some motion data are lost.

As prevention,

1. The host requires to response to motion interrupt triggering as soon as possible to read motion data register.
2. Optimize the chip resolution.

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7.4.6 Raw Data Output

This section describes the method to output a full array of raw data values of a single frame.

In order to enable the Raw Data Output, write *Raw_Data_Output* with value 0x81, followed by polling the same *Raw_Data_Output* register until *RD_VALID* is read as 1, which indicates *Raw_Data_Output* is ready and can be read out serially via the *Raw_Data_Output* register.

7.4.6.1 Raw Data Output Procedure

The chip must be initialized and reset correctly.

1. Write register 0x7F with value 0x00
2. Write register 0x55 with value 0x04
3. Write register 0x40 with value 0x80
4. Read register 0x02 (Motion[1:0]) to confirm *OP_State* [1:0] equals to “00”.
5. Write register 0x1A with value 0x81
6. Continuously read register 0x1A until *RD_VALID* is read as “1”.
7. Burst read register 0x1B with burst length of 1296 bytes to form a complete picture element array.
8. Write register 0x40 with value 0x00
9. Write register 0x55 with value 0x00

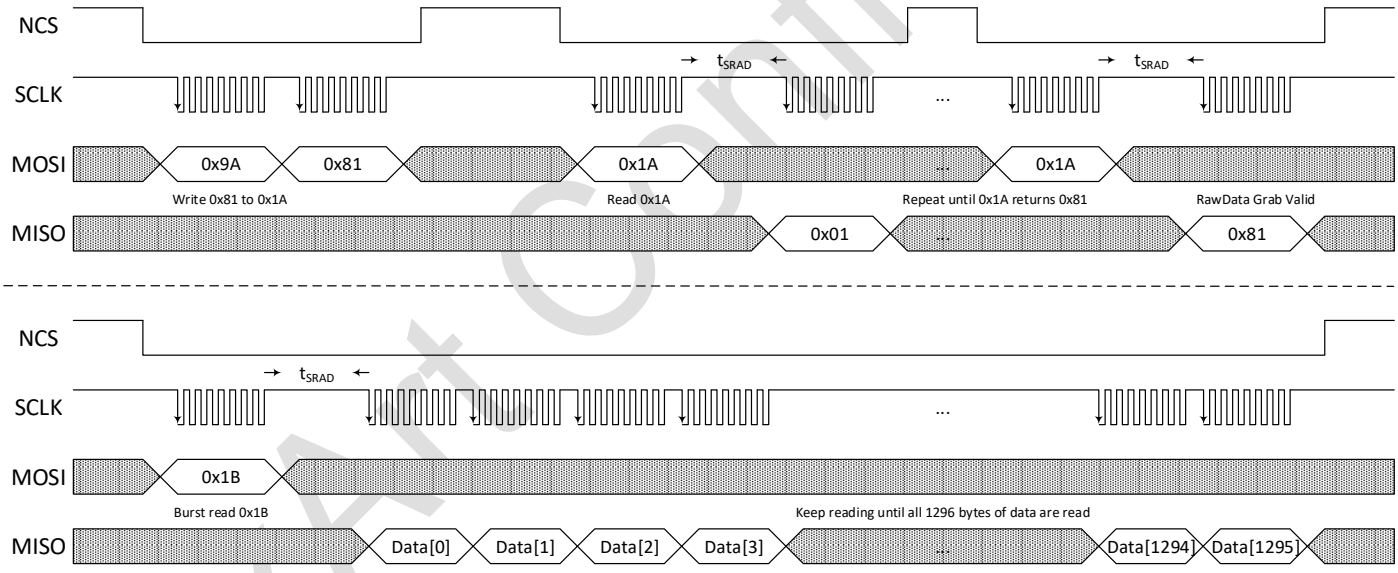
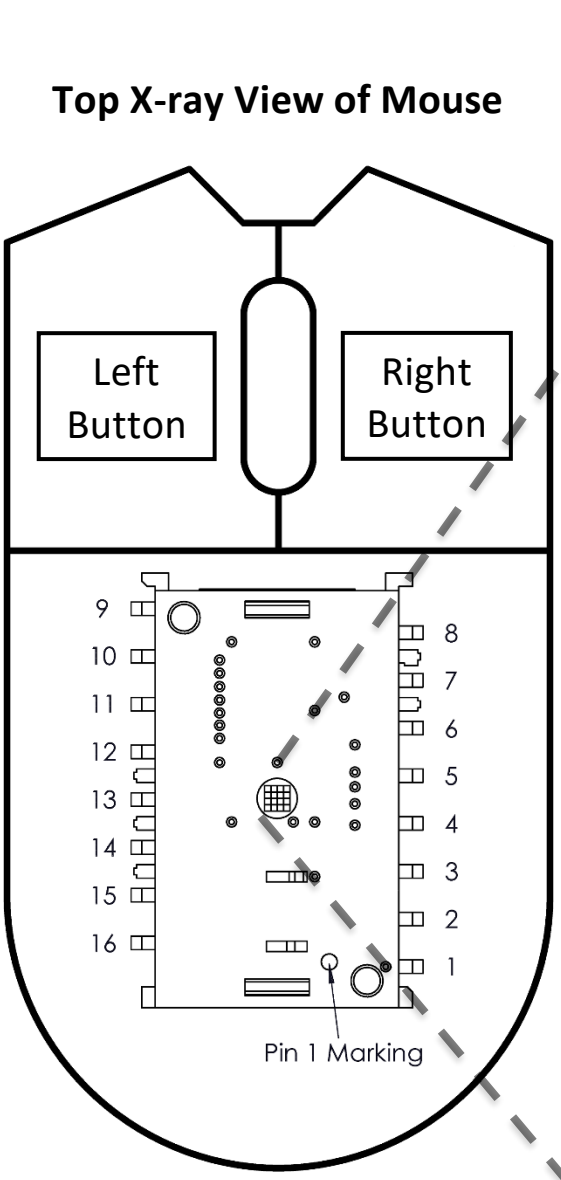


Figure 27. Raw Data Output Waveform

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7.4.6.2 Raw Data Output Mapping



35	71	107		1187	1223	1259	1295
34	70	106		1186	1222	1258	1294
33	69	105		1185	1221	1257	1293
32	68	104		1184	1220	1256	1292
31	67	103		.	1219	1255	1291
30	66	102		.	1218	1254	1290
29	65	101		.	1217	1253	1289
28	64	100		etc	1216	1252	1288
27	63	99		↑	1215	1251	1287
26	62	98			1214	1250	1286
25	61	97			1213	1249	1285
24	60	96			1212	1248	1284
23	59	95			1211	1247	1283
22	58	94			1210	1246	1282
21	57	93			1209	1245	1281
20	56	92			1208	1244	1280
19	55	91			1207	1243	1279
18	54	90			1206	1242	1278
17	53	89			1205	1241	1277
16	52	88			1204	1240	1276
15	51	87			1203	1239	1275
14	50	86			1202	1238	1274
13	49	85			1201	1237	1273
12	48	84			1200	1236	1272
11	47	83			1199	1235	1271
10	46	82			1198	1234	1270
9	45	81			1197	1233	1269
8	44	80	↑		1196	1232	1268
7	43	79	etc		1195	1231	1267
6	42	78	.		1194	1230	1266
5	41	77	.		1193	1229	1265
4	40	76	.		1192	1228	1264
3	39	75	111		1191	1227	1263
2	38	74	110		1190	1226	1262
1	37	73	109		1189	1225	1261
0	36	72	108		1188	1224	1260

Figure 28. Raw Data Map (Surface Referenced)

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7.4.7 Related Registers

Usage	Name	Bank	Address
Output Access	*Motion*	0x00	0x02
	Delta_X_L	0x00	0x03
	Delta_X_H	0x00	0x04
	Delta_Y_L	0x00	0x05
	Delta_Y_H	0x00	0x06
Burst Read Operation	*Motion_Burst*	0x00	0x16
Raw Data Output	*Raw_Data_Output_Status*	0x00	0x1A
	Raw_Data_Output	0x00	0x1B

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7.5 Motion Sync

Motion sync is a feature embedded in the chip that provides smoother motion report over time plot as shown in the following figures.

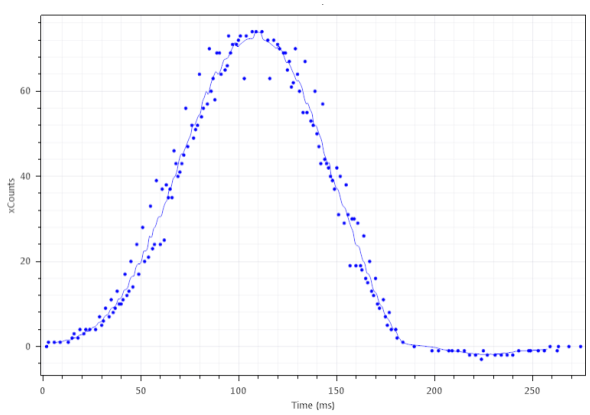


Figure 29. Motion Report vs Time Plot (Without Motion Sync)

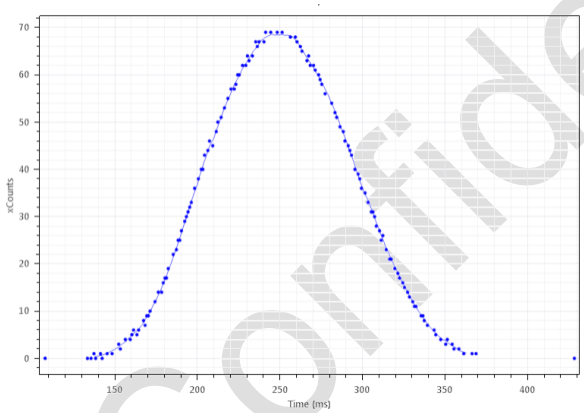


Figure 30. Motion Report vs Time Plot (With Motion Sync)

7.5.1 Motion Sync Setting

The Motion Sync is enabled by default after loaded the initialization settings. It can be disabled or enabled by writing the following register commands.

Write the following register values to disable Motion Sync:

1. Write register 0x7F with value 0x0D
2. Write register 0x48 with value 0xFC
3. Write register 0x7F with value 0x00

Write the following register values to enable Motion Sync:

1. Write register 0x7F with value 0x0D
2. Write register 0x48 with value 0xFD
3. Write register 0x7F with value 0x00

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7.5.2 Implementation Requirement

7.5.2.1 Enable Motion Sync

1. When implementing the Motion Sync, the MCU must execute polling interval at a fixed and consistent interval whenever the MOTION pin is active (LOW).
2. Different polling rate must be applied for different gaming modes, as indicated in the table below. It is important to note that the polling rate must be carried out at consistent interval with timing tolerance of ± 2% (with respect to polling rate).

Operation Mode		Max. Polling Rate
Gaming Mode	High Performance	1kHz
	Low Power	1kHz
	Corded	8kHz
Office Mode		125Hz
Glass Mode		1kHz

3. When polling the MOTION pin, the MCU must poll the motion data (DeltaX and DeltaY) twice after the MOTION pin goes LOW. Below is the sample Code of polling the MOTION pin:

```
if (CHECK_MOTION_PIN() == LOW )      // Read MOTION PIN
{
    read_count = 2;                  // set counter = 2
}
if(poll_usb && (read_count > 0))     // 1ms polling
{
    nsAPI_SSP_Sensor_Monitor();      // busrt read (read Delta X and Delta Y)
    nsAPI_Main_MouseEventHandle();   // USB report
    poll_usb = 0;
    read_count -- ;                  // decrement of read_count
}
```

7.5.2.2 Disable Motion Sync

When Motion Sync is disabled, the MOTION pin goes LOW whenever there is non-zero data in the Delta_X_L, Delta_X_H, Delta_Y_L, or Delta_Y_H registers. After reading the motion data, the MOTION pin goes HIGH.

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7.6 Lift Cut-off

The chip provides three universal lift cut-off settings (A, B and C) to adjust the lift cut-off height. The settings are applicable to all navigation surfaces. *Lift_Config* is used to configure the universal lift cut-off height setting. Upon the chip initialization, the chip is configured to lift setting A as default.

The chip also provides manual lift cut-off calibration to optimize lift performance by tuning parameters on a dedicated surface.

7.6.1 Manual Lift Cut-off Calibration

The chip provides manual lift cut-off calibration to optimize lift performance by tuning parameters on a dedicated surface. This feature involves user interaction.

7.6.2 Calibration Procedure

1. Ensure the chip is powered up according to the System Initialization in Section 7.1.

2. Prompt the user for manual lift cut-off calibration is about to begin and ensure the mouse is placed on the surface (mouse is not lifted).

3. Start the calibration procedure by loading the following register commands.
   a. Read register 0x40 and store its value into Var_Mode
   b. Write register 0x7F with value 0x00
   c. Write register 0x40 with value 0x80
   d. Write register 0x7F with value 0x18
   e. Write register 0x47 with value 0x27
   f. Write register 0x7A with value 0x00
   g. Write register 0x41 with value 0x01
   h. Write register 0x43 with value 0x01
   i. Write register 0x59 with value 0xA4

4. The calibration procedure can be started with a software prompt to the user and user-initiated through mouse-click event. Recommend to move the mouse over a distance of minimum 20 inch to cover most of the area on the surface.

5. Calibration can proceed to the next step or continue until user initiates a mouse-click event.

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6. Read and store the calibrated values, write the following register commands to stop the Calibration.
   a. Read register 0x7B and store its value into VarA
   b. Read register 0x7C and store its value into VarB
   c. Read register 0x7D and store its value into VarC
   d. Wait for 1ms
   e. Write register 0x7F with value 0x18
   f. Write register 0x59 with value 0xA6
   g. Write register 0x47 with value 0x21
   h. Write register 0x40 with value 0x00
   i. Write register 0x7F with value 0x00
   j. Write register 0x40 with Var_Mode

7. Prompt the user the calibration process is completed. The three calibrated values, VarA, VarB and VarC will be used to enable the calibration setting in the next section.

7.6.3 Enable Manual Lift Cut-off Calibration Setting

Write the following register commands to enable the manual calibration setting on the calibrated surface.

1. Read register 0x40 and store its value into a variable “Var_Mode”
2. Write register 0x40 with value 0x80
3. Continuously read register 0x02 until both bit-0 and bit-1 is read as “0”
4. Write register 0x7F with value 0x18
5. Write register 0x7B with VarA
6. Write register 0x7C with VarB
7. Write register 0x7D with VarC
8. Write register 0x7A with value 0x00
9. Write register 0x41 with value 0x01
10. Write register 0x43 with value 0x01
11. Write register 0x59 with value 0xA4
12. Write register 0x47 with value 0x21
13. Write register 0x40 with value 0x00
14. Wait for 10ms
15. Write register 0x47 with value 0x23
16. Write register 0x7F with value 0x00
17. Write register 0x40 with Var_Mode

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7.6.4 Disable Lift Cut-off Calibration Register Setting

Write the following register commands to disable manual calibration setting and configure the chip to universal lift cut-off Setting A.

1. Write register 0x7F with value 0x18
2. Write register 0x40 with value 0x83
3. Write register 0x41 with value 0x02
4. Write register 0x43 with value 0x80
5. Write register 0x59 with value 0xA6
6. Write register 0x7A with value 0x00
7. Write register 0x47 with value 0x21
8. Write register 0x7F with value 0x00

7.6.5 Related Register

Usage	Register Name	Bank	Address
Lift Cut-off	*Lift_Config*	0x18	0x7A

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7.7 Settings

7.7.1 Resolution

RESOLUTION_X and *RESOLUTION_Y* registers configures the resolution of the X-axis and Y-axis respectively. After update the resolution setting, either in *RESOLUTION_X* or/and *RESOLUTION_Y*, write value 0x01 into *SET_RESOLUTION* for the chip to use the new resolution setting.

7.7.2 Angle Snap

Angle Snap for Gaming Mode is disabled by default after chip initialization, it can be enabled or disabled through the *Angle_Snap* register.

7.7.3 Angle Tune

Angle Tune is a function to perform delta report rotation. When it is enabled, the chip transforms the delta report from 0 degree to the defined rotational angle *in Angle_Tune1* register. Angle Tune is disabled by default after chip initialization, it can be enabled or disabled in *Angle_Tune2* register.

7.7.4 Related Register

Usage	Register Name	Bank	Address
Resolution	*Resolution_X_Low*	0x00	0x48
	Resolution_X_High		0x49
	Resolution_Y_Low		0x4A
	Resolution_Y_High		0x4B
	Set_Resoltuion		0x47
Angle Snap	*Angle_Snap*	0x00	0x56
Angle Tune	*Angle_Tune1*	0x05	0x77
	Angle_Tune2		0x78

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8.0 Registers

8.1 Registers Map

Table 17. Non-Bank Register

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x7F	-	Bank_Switch	R/W	-	0x00

Table 18. Register Bank 0x00

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x00	-	Product_ID	R	-	0x53
0x01	-	Revision_ID	R	-	0x00
0x02	-	Motion	R/W	-	0x00
	Bit[7]	MOT		0	
	Bit[3]	Lift_Stat		0	
	Bit[2]	OP_Mode		0	
	Bit[1:0]	OP_State		0	
0x03	-	Delta_X_L	R	-	0x00
0x04	-	Delta_X_H	R	-	0x00
0x05	-	Delta_Y_L	R	-	0x00
0x06	-	Delta_Y_H	R	-	0x00
0x07	-	SQUAL	R	-	0x00
0x08	-	Raw_Data_Sum	R	-	0x00
0x09	-	Maximum_Raw_Data	R	-	0x00
0x0A	-	Minimum_Raw_Data	R	-	0x00
0x0B	-	Shutter_Lower	R	-	0x00
0x0C	-	Shutter_Upper	R	-	0x00
0x15	-	Observation	R/W	-	0x00
0x16	-	Motion_Burst	R	-	0x00
0x1A	-	Raw_Data_Output_Status	R/W	-	0x00
	Bit[7]	RD_VALID		0	
	Bit[0]	RD_OUTPUT_EN		0	
0x1B	-	Raw_Data_Output	R	-	0x00
0x3A	-	Power_Up_Reset	W	-	N/A
0x3B	-	Shutdown	W	-	N/A
0x40	-	Performance	R/W	-	0x00
	Bit[7]	AWAKE		0	
0x47	-	Set_Resolution	W	-	N/A
0x48	-	Resolution_X_Low	R/W	-	0x63
0x49	-	Resolution_X_High	R/W	-	0x00

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Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x4A	-	Resolution_Y_Low	R/W	-	0x63
0x4B	-	Resolution_Y_High	R/W	-	0x00
0x56	-	Angle_Snap	R/W	-	0x00
0x5B	-	Axis_Control	R/W	-	0x60
	Bit[7]	SWAP_XY		0	
	Bit[6]	INV_Y		1	
	Bit[5]	INV_X		1	
0x5C	-	Motion_Control	R/W	-	0x22
	Bit[7]	MOT_SET		0	
	Bit[1]	RES_MOD		1	
0x5F	-	Inverse_Product_ID	R	-	0xAC
0x74	-	Glass_Rest1_Period	R/W	-	0x05
0x75	-	Glass_Rest1_Downshift	R/W	-	0x20
0x77	-	Run_Downshift	R/W	-	0x14
0x78	-	Rest1_Period	R/W	-	0x01
0x79	-	Rest1_Downshift	R/W	-	0x9C
0x7A	-	Rest2_Period	R/W	-	0x19
0x7B	-	Rest2_Downshift	R/W	-	0x5E
0x7C	-	Rest3_Period	R/W	-	0x3F
0x7D	-	Run_Downshift_Mult	R/W	-	0x07
	Bit[3:0]	RUN_M		7	
0x7E	-	Rest_Downshift_Mult	R/W	-	0x55
	Bit[6:4]	REST2_M		5	
	Bit[2:0]	REST1_M		5	

Note: Switch to Register Bank 0x00 by writing 0x00 to register 0x7F.

Table 19. Register Bank 0x05

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x77	-	Angle_Tune1	R/W	-	0x00
0x78	-	Angle_Tune2	R/W	-	0x00

Note: Switch to Register Bank 0x05 by writing 0x05 to register 0x7F.

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Table 20. Register Bank 0x0D

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x4F	-	Ripple Control	R/W	-	0x80

Note: Switch to Register Bank 0x0D by writing 0x0D to register 0x7F.

Table 21. Register Bank 0x18

Address	Bit Field	Name	Access	Bit Field Default Value	Register Default Value
0x7A	-	Lift_Config	R/W	-	0x85

Note: Switch to Register Bank 0x18 by writing 0x18 to register 0x7F.

8.2 Register Description

8.2.1 Product ID

Register Name	Product_ID		
Bank	0x00	**Address**	0x00
Access	R	**Default Value**	0x53
Description	This value is a unique identification assigned to this model only. The value in this register does not change; it can be used to verify that the serial communications link is functional.		

Register Name	Revision_ID		
Bank	0x00	**Address**	0x01
Access	R	**Default Value**	0x00
Description	This register contains the current IC revision. It is subject to change when new IC versions are released.		

Register Name	Inverse_Product_ID		
Bank	0x00	**Address**	0x5F
Access	R	**Default Value**	0xAC
Description	This value is the inverse of the *Product_ID*. It can be used to test the SPI port hardware.		

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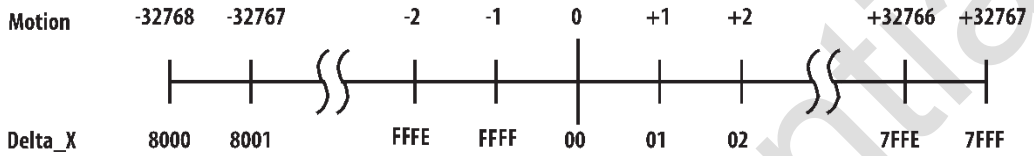
8.2.2 Motion

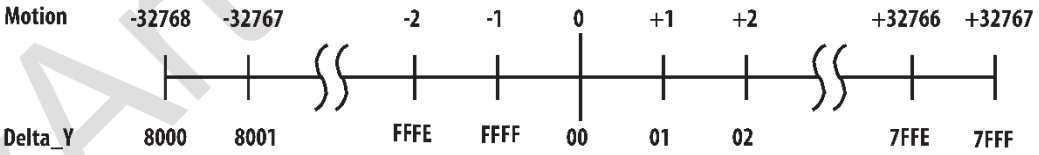
Register Name	Motion		
Bank	0x00	**Address**	0x02
Access	R/W	**Default Value**	0x00
Description	This register allows the user to determine if motion has occurred since the last time it was read. The procedure to read the motion registers (*Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H*) is as follows: 1. Read the Motion register. This will freeze the *Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H* register values. 2. If the MOT bit is set, *Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H* registers must be read in the given sequence to get the accumulated motion. **Note:** if *Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H* registers are not read before the motion register is read for the second time, the data in *Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H* will be lost. 3. To read a new set of motion data (*Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H*), repeat from Step 1. **Note:** Write any value to this register will clear all the motion data.		

Bit Field	Name	Default Value	Description
7	MOT	0	Motion since last report 0: No motion 1: Motion occurred, data in *Delta_X_L, Delta_X_H, Delta_Y_L* and *Delta_Y_H* registers is ready to be read.
5	Reset_Flag	0	Indicate the reset status of the chip. 0: After Reset Event 1: After chip initialization in the section 7.1.1. **Note:** Reset_Flag can be used as part of the recovery scheme to detect if a reset is triggered (Reset_Flag = 0) unintentionally caused by a reset command, full chip reset or ESD event. The Reset_Flag will be set to 1 after the chip start-up per the recommended Initialization Register Setting in the section 7.1.2.
3	Lift_Stat	0	Indicate the lift status of the chip 0: Chip on surface 1: Chip lifted
2	OP_Mode	0	0: Glass Mode 1: Gaming Mode and Office Mode
1:0	OP_State	0	'00': Run State '01': Rest 1 State '10': Rest 2 State '11': Rest 3 State

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Register Name	Delta_X_L		
Bank	0x00	Address	0x03
Access	R	Default Value	0x00
Register Name	**Delta_X_H**		
Bank	0x00	Address	0x04
Access	R	Default Value	0x00
Description	16-bit 2's complement number. Lower 8 bits (*Delta_X_L*) and Upper 8 bits (*Delta_X_H*) of Delta_X. X movement counts since last report. Absolute value is determined by resolution. *Delta_X_H* must be read after *Delta_X_L* to have the full motion data.  Note: It is recommended that registers 0x02, 0x03, 0x04, 0x05 and 0x06 be read sequentially.		

Register Name	Delta_Y_L		
Bank	0x00	Address	0x05
Access	R	Default Value	0x00
Register Name	**Delta_Y_H**		
Bank	0x00	Address	0x06
Access	R	Default Value	0x00
Description	16-bit 2's complement number. Lower 8 bits (*Delta_Y_L*) and Upper 8 bits (*Delta_Y_H*) of Delta_Y. Y movement counts since last report. Absolute value is determined by resolution. *Delta_Y_H* must be read after *Delta_Y_L* to have the full motion data.  Note: It is recommended that registers 0x02, 0x03, 0x04, 0x05 and 0x06 to be read sequentially.		

Register Name	Motion_Burst		
Bank	0x00	Address	0x16
Access	R	Default Value	0x00
Description	The *Motion_Burst* register is used for high-speed access of up to 12 register bytes. Refer to Section 7.4.4 for more details.		

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Register Name	Raw_Data_Output_Status		
Bank	0x00	**Address**	0x1A
Access	R	**Default Value**	0x00
Description	This register allows user to enable Raw Data Output and poll for the data valid status. Refer to section 7.4.6 for the detail of the Raw Data Output process.		
Bit Field	**Name**	**Default Value**	**Description**
7	RD_VALID	0	0: Raw Data Output not valid 1: Raw Data Output valid
6:1	Reserved	0	Reserved
0	RD_OUTPUT_EN	0	0: Disable Raw Data Output 1: Enable Raw Data Output

Register Name	Raw_Data_Output		
Bank	0x00	**Address**	0x1B
Access	R/W	**Default Value**	0x00
Description	This register contains the raw data values when Raw Data Output process is enabled. Refer to section 7.4.6 for the detail of the Raw Data Output process.		

8.2.3 Operational Control

Register Name	Performance		
Bank	0x00	**Address**	0x40
Access	R/W	**Default Value**	0x00
Description	This register configures the Rest State of the chip.		
Bit Field	**Name**	**Default Value**	**Description**
7	AWAKE	0	0: Enable Rest State 1: Disable Rest State
6:0	Reserved	0	Reserved

Register Name	Set_Resolution		
Bank	0x00	**Address**	0x47
Access	W	**Default Value**	0x00
Description	After update the resolution setting, either in *RESOLUTION_X* or/and *RESOLUTION_Y*, write value 0x01 into *SET_RESOLUTION* for the chip to use the new resolution setting. 0x01: update resolution setting		

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Register Name	Resolution_X_Low		
Bank	0x00	**Address**	0x48
Access	R/W	**Default Value**	0x63
Register Name	**Resolution_X_High**		
Bank	0x00	**Address**	0x49
Access	R/W	**Default Value**	0x00
Description	Lower byte and upper byte of RESOLUION_X register. This register allows to change the resolution of X-axis of the chip. Write to *RESOLUTION_X_LOW* first, then *RESOLUTION_X_HIGH*. They must be written consecutively. After update the resolution setting, either in *RESOLUTION_X* or/and *RESOLUTION_Y*, write value 0x01 into *SET_RESOLUTION* for the chip to use the new resolution setting. Set X-axis Resolution: 0x0000: 50CPI 0x0001: 100CPI 0x0002: 150CPI : 0x0063: 5000CPI : 0x018F: 20000CPI : 0x0207: 26000CPI : 0x0257: 30000CPI (max.)		

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Register Name	Resolution_Y_Low		
Bank	0x00	**Address**	0x4A
Access	R/W	**Default Value**	0x63
Register Name	**Resolution_Y_High**		
Bank	0x00	**Address**	0x4B
Access	R/W	**Default Value**	0x00
Description	Lower byte and upper byte of RESOLUION_Y register. This register allows to change the resolution of Y-axis of the chip. Write to *RESOLUTION_Y_LOW* first, then *RESOLUTION_Y_HIGH*. They must be written consecutively. After update the resolution setting, either in *RESOLUTION_X* or/and *RESOLUTION_Y*, write value 0x01 into *SET_RESOLUTION* for the chip to use the new resolution setting. Set Y-axis Resolution: 0x0000: 50CPI 0x0001: 100CPI 0x0002: 150CPI : 0x0063: 5000CPI : 0x018F: 20000CPI : 0x0207: 26000CPI : 0x0257: 30000CPI (max.)		

Register Name	Angle_Snap		
Bank	0x00	**Address**	0x56
Access	R/W	**Default Value**	0x00
Description	Write 0x0D to this register to enable Angle Snap feature for Gaming Mode.		

Register Name	Axis_Control		
Bank	0x00	**Address**	0x5B
Access	R/W	**Default Value**	0x60
Description	The register set the axis direction of the chip reporting.		

Bit Field	Name	Default Value	Description
7	SWAP_XY	0	1: Swap X/Y directions
6	INV_Y	1	1: Invert Y direction
5	INV_X	1	1: Invert X direction
4:0	Reserved	0	Reserved

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Register Name	Motion_Control		
Bank	0x00	**Address**	0x5C
Access	R/W	**Default Value**	0x22
Description	Configures the motion pin setting and select the X-axis and Y-axis resolution mode.		

Bit Field	Name	Default Value	Description
7	MOT_SET	0	0: Motion pin active low 1: Motion pin active high
6	Reserved	0	Reserved
5	Reserved	1	Reserved
4:2	Reserved	0	Reserved
1	RES_MOD	1	0: Both X-axis and Y-axis resolution are defined by *RESOLUTION_X_LOW* and *RESOLUTION_X_HIGH* 1: X-axis resolution is defined by *RESOLUTION_X_LOW* and *RESOLUTION_X_HIGH* and Y-axis resolution is defined by *RESOLUION_Y_LOW* and *RESOLUION_Y_HIGH* (default)
0	Reserved	0	Reserved

Register Name	Lift_Config		
Bank	0x18	**Address**	0x7A
Access	R/W	**Default Value**	0x85
Description	Write this register to configure lift setting. Upon the chip initialization, the chip is configured to lift setting A as default. **Value** — **Lift Setting** 0x00 — Lift Setting A 0x85 — Lift Setting B 0x8F — Lift Setting C Other value — Reserved		

Register Name	Ripple_Control		
Bank	0x0D	**Address**	0x4F
Access	R/W	**Default Value**	0x80
Description	Write to this register to enable or disable the Ripple Control feature. Upon chip start-up, as per the recommended initialization sequence, Ripple Control is disabled by default. 0x80: Disable Ripple Control 0x82: Enable Ripple Control		

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Register Name	Angle_Tune1		
Bank	0x05	**Address**	0x77
Access	R/W	**Default Value**	0x00
Description	Angle Tune is a function to perform delta report rotation. When enabled, the chip able to transform the delta report from 0 degree to the defined rotational angle in this register. Write 0x80 to *ANGLE_TUNE2* register to enable it. Each step in the register is equivalent to approximate 1 degree. The setting for negative degree is computed by using 2's complement value of the set degree. 0xE2: -30 degree 0xF6: -10 degree 0x00: 0 degree (default) 0x0F: +15 degree 0x1E: +30 degree		

Register Name	Angle_Tune2		
Bank	0x05	**Address**	0x78
Access	R/W	**Default Value**	0x00
Description	Write to this register to enable Angle Tune.		

Bit Field	Name	Default Value	Description
7	EN	0	0: Disable Angle Tune 1: Enable Angle Tune
6:0	Reserved	0	Reserved

8.2.4 Operational Check

Register Name	SQUAL		
Bank	0x00	**Address**	0x07
Access	R	**Default Value**	0x00
Description	The SQUAL (Surface Quality) register is a measure of the number of valid features visible by the chip in the current frame. Use the following formula to find the total number of valid features. Number of Features = *SQUAL* Register Value x 4 The maximum SQUAL register value is 0xFF. Since small changes in the current frame can result in changes in SQUAL, variations in SQUAL when looking at a surface is expected. SQUAL values are only valid when chip is in Run State. Disable Rest State before reading SQUAL.		

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Register Name	Raw_Data_Sum		
Bank	0x00	**Address**	0x08
Access	R	**Default Value**	0x00
Description	This register is used to find the chip average raw data value. It reports the upper byte of an 18-bit counter which sums all 1296 raw data in the current frame. To find the average raw data value follows the formula below. Gaming Mode: Average pixel value= Raw_Data_Sum x 1024/1296 Glass Mode: Average pixel value= Raw_Data_Sum x 1024 x 2/1296 The maximum register value is 0xA0 (hex) or 160 (dec) and the minimum register value is 0. The Raw_Data_Sum value can change every frame. Disable rest state before reading Raw_Data_Sum value.		

Register Name	Maximum_Raw_Data		
Bank	0x00	**Address**	0x09
Access	R	**Default Value**	0x00
Description	Maximum raw data value in current frame. Minimum value = 0, maximum value = 255. The maximum raw data value can be changed in every frame.		

Register Name	Minimum_Raw_Data		
Bank	0x00	**Address**	0x0A
Access	R	**Default Value**	0x00
Description	Minimum raw data value in current frame. Minimum value = 0, maximum value = 255. The minimum raw data value can be changed in every frame.		

Register Name	Shutter_Lower		
Bank	0x00	**Address**	0x0B
Access	R	**Default Value**	0x00
Register Name	**Shutter_Upper**		
Bank	0x00	**Address**	0x0C
Access	R	**Default Value**	0x00
Description	Lower byte and upper 5-bit of the 13-bit Shutter register. Units are clock cycles of the internal oscillator. Read *Shutter_Upper* first, then *Shutter_Lower*. They must be read consecutively. The shutter is adjusted to keep the average raw data values within normal operating ranges. The shutter value is checked and automatically adjusted to a new value if needed on every frame.		

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8.2.5 Troubleshooting

Register Name	Observation		
Bank	0x00	**Address**	0x15
Access	R/W	**Default Value**	0x80
Description	The user must clear the register by writing 0x00, wait for a minimum t_{dly_obs} ms and read the register. If the chip is working correctly, the value of bit[7:0] must be **0xB7** or **0xBF.** The register can be used as part of recovery scheme to detect a problem caused by EFT/B or ESD event. t_{dly_obs} is defined as the longest frame period + 10% variation. The longest frame period is when chip is in Rest3 state. Clock frequency tolerance value need to be considered. For example, if the default Rest3 period of 500ms is used, then t_{dly_obs} = 500ms + 50ms		

8.2.6 Power Management

Register Name	Glass_Rest1_Period		
Bank	0x00	**Address**	0x74
Access	R/W	**Default Value**	0x05
Description	This register set the Glass Rest1 period. Glass Rest1 period = *Glass_Rest1_Period[7:0]* x 1ms Upon the chip start-up per the recommended Power-on Sequence, Glass Rest1 period is set to 5ms as default. Default Glass Rest1 period = (0x05) x 1ms = 5 x 1ms = 5ms Minimum value is 0x01. A value of 0x00 is invalid. All the above values are expected to have ± 10% tolerance.		

Register Name	Glass_Rest1_Downshift		
Bank	0x00	**Address**	0x75
Access	R/W	**Default Value**	0x20
Description	This register set the Glass Rest1 to Rest2 downshift time. Use the formula below for calculation. Glass Rest1 Downshift time (ms) = *Glass_Rest1_Downshift[7:0]* x *Rest_downshift_Mult*[2:0] (default 64) x Glass_*Rest1*_period (default 5ms) Upon the chip start-up per the recommended Power-on Sequence, Glass Rest1 Downshift time is set to 10s as default. Default Glass Rest1 Downshift time = (0x20) x 64 x 5ms = 32 x 64 x 5ms = 10240ms = 10.24s Minimum value is 0x01. A value of 0x00 will be internally clipped to 0x01. All the above values are expected to have ± 10% tolerance.		

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Register Name	Run_Downshift		
Bank	0x00	**Address**	0x77
Access	R/W	**Default Value**	0x14
Description	This register set the Run to Rest1 downshift time. Use the formula below for calculation. Run Downshift time (ms) = *Run_Downshift*[7:0] x *Run_Downshift_Mult*[3:0] (default 256) x 50µs Upon the chip start-up per the recommended Power-on Sequence, Run Downshift time is set to 1s as default. Default Run Downshift time = (0x4F) x 256 x 50µs = 79 x 256 x 50µs = 1s Minimum value is 0x01. A value of 0x00 will be internally clipped to 0x01. All the above values are expected to have ± 10% tolerance.		

Register Name	Rest1_Period		
Bank	0x00	**Address**	0x78
Access	R/W	**Default Value**	0x01
Description	This register set the Rest1 period. Rest1 period = *Rest1_Period[7:0]* x 1ms Upon the chip start-up per the recommended Power-Up Sequence, Rest1 period is set to 1ms as default. Default Rest1 period = (0x01) x 1ms = 1 x 1ms = 1ms Minimum value is 0x01. A value of 0x00 is invalid. All the above values are expected to have ± 10% tolerance.		

Register Name	Rest1_Downshift		
Bank	0x00	**Address**	0x79
Access	R/W	**Default Value**	0x9C
Description	This register set the Rest1 to Rest2 downshift time. Use the formula below for calculation. Rest1 Downshift time (ms) = *Rest1_Downshift*[7:0] x *Rest_Downshift_Mult*[2:0] (default 64) x *Rest1_period* (default 1ms) Upon the chip start-up per the recommended Power-Up Sequence, Rest1 Downshift *time* is set to 10s as default. Default Rest 1 Downshift time = (0x9C) x 64 x 1ms = 156 x 64 x 1ms = 9984ms = 10s Minimum value is 0x01. A value of 0x00 will be internally clipped to 0x01. All the above values are expected to have ± 10% tolerance.		

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Register Name	Rest2_Period		
Bank	0x00	**Address**	0x7A
Access	R/W	**Default Value**	0x19
Description	This register set the Rest2 period. Rest2 period = *Rest2_Period*[7:0] x 1ms x 4 Upon the chip start-up per the recommended Power-on Sequence, *Rest2 period* is set to 100ms as default. Default Rest2 period = (0x19) x 1ms x 4 = 25 x 1ms x 4 = 100ms Minimum value is 0x01. A value of 0x00 is invalid. All the above values are expected to have ± 10% tolerance.		

Register Name	Rest2_Downshift		
Bank	0x00	**Address**	0x7B
Access	R/W	**Default Value**	0x5E
Description	This register set the Rest2 to Rest3 downshift time. Use the formula below for calculation. Rest2 Downshift time (ms) = *Rest2_Downshit*[7:0] x *Rest_Downshift_Mult*[6:4] (default 64) x *rest2_period* (default 100ms) Upon the chip start-up per the recommended Power-on Sequence, Rest2 Downshift time is set to 10 minutes as default. Default Rest2 Downshift time = (0x5E) x 64 x 100ms = 94 x 64 x 100ms = 601.6s = 10min Minimum value is 0x01. A value of 0x00 will be internally clipped to 0x01. All the above values are expected to have ± 10% tolerance.		

Register Name	Rest3_Period		
Bank	0x00	**Address**	0x7C
Access	R/W	**Default Value**	0x3F
Description	This register set the Rest3 period. Rest3 period = *Rest3_Period*[7:0] x 1ms x 8 Upon the chip start-up per the recommended Power-on Sequence, *Rest3_period* is set to 504ms as default. Default Rest3 period = (0x3F) x 1ms x 8 = 63 x 1ms x 8 = 504ms Minimum value is 0x01. A value of 0x00 is invalid. All the above values are expected to have ± 10% tolerance.		

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Register Name	Run_Downshift_Mult		
Bank	0x00	**Address**	0x7D
Access	R/W	**Default Value**	0x07
Description	This register set the RUN Downshift Multiplier. Refer to the formula in Register *Run_Downshift* for the detail.		

Bit Field	Name	Default Value	Description
7:4	Reserved	0	Reserved
3:0	RUN_M	7	**Value** **Run_Mult** 0 2 1 4 2 8 3 16 4 32 5 64 6 128 7 256 8 512 9 1024 A 2048

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Register Name	Rest_Downshift_Mult			
Bank	0x00		**Address**	0x7E
Access	R/W		**Default Value**	0x55
Description	This register set the REST Downshift Multiplier. Refer to the formula in Register *Rest1_Downshift* and *Rest2_Downshift* for the use detail.			
Bit Field	**Name**	**Default Value**	**Description**	
7	Reserved	0	Reserved	
6:4	REST2_M	5	**Value** / **REST2_M** 0 / 2 1 / 4 2 / 8 3 / 16 4 / 32 5 / 64 6 / 128 7 / 256	
3	Reserved	0	Reserved	
2:0	REST1_M	5	**Value** / **REST1_M** 0 / 2 1 / 4 2 / 8 3 / 16 4 / 32 5 / 64 6 / 128 7 / 256	

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8.2.7 Reset

- Shutdown Register

Register Name	Shutdown		
Bank	0x00	**Address**	0x3B
Access	W	**Default Value**	N/A
Description	Shutdown state control register. Write register 0x3B with value to enter Shutdown state. Refer to section 5.3.3 for the detail.		

- Reset Register

Register Name	Power_Up_Reset		
Bank	0x00	**Address**	0x3A
Access	W	**Default Value**	N/A
Description	Write 0x5A to this register to reset the chip. All settings will revert to default values.		

8.2.8 Bank Switch

Register Name	Bank_Switch		
Bank	NA	**Address**	0x7F
Access	W/R	**Default Value**	0x00
Description	Write this register to switch the register bank.		

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Revision History

Revision Number	Date	Description
1.0	04 Dec 2023	Initial Release
1.1	29 Feb 2024	Section 7.1.2: Update Initialization Setting Section 7.2.4: Update Corded Gaming and Glass Mode (< 7000cpi) Setting Section 7.2.5: Update Corded Gaming and Glass Mode (> 7000cpi) Setting Section 7.6.3: Update “Enable Manual Lift Cut-off Calibration Setting” Section 8.2.2: Include Reset_Flag in the Motion Register Section 7.5: new section to include “Motion Sync”
1.2	18 June 2024	Section 7.2: Update all the mode settings Section 7.6.4: Update “Disable Lift Cut-off Calibration Register Setting”

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